A Pre-Registered Check of the Off-Policy Evaluation Benchmark on the Open Bandit Dataset

Technical report

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Abstract

The Open Bandit Dataset is used to test off-policy estimators on real logs: the click rate of a Bernoulli Thompson sampling (BTS) policy is estimated from uniform-random logs and compared with the click rate of the BTS logs, the on-policy value. On the Men's campaign the inverse propensity score (IPS) estimate is 0.891 of the on-policy value, a number known from earlier work, and a published bootstrap interval over rows excludes the on-policy value. This report asks how the estimate changes when the evaluation target, a published prior, is aligned with what the BTS log shows. Propositions and decision rules were committed to the study's repository 79 minutes before they were applied, in one run, to the other two campaigns, All and Women's; the outcome on Men's and public point estimates for the other two were known at that time. On All and Women's the distribution implied by the prior differs from the hourly frequencies of the BTS log by a total variation distance of 0.526 and 0.527, against noise floors of 0.136 and 0.115, which quantifies a mismatch that a public issue had reported. No shortfall is shown: the IPS ratio is 0.948 (0.867 to 1.033), 1.004 (0.917 to 1.099) and, pooled, 0.974 (0.915 to 1.036), with intervals from a bootstrap over ten-minute blocks. With the target aligned to the BTS log by three-hour period the estimate is 1.045 (0.978 to 1.114) of the on-policy value pooled, a paired change of 0.072 (0.002 to 0.139) whose interval excludes 0 in the pooled unit only; under the registered rules the gap is not read where no shortfall is established, so the primary question has no reading. Post hoc, the standard deviation of the ratio over block replicates is 1.31 to 1.47 times the one over rows or impressions; on Men's the intervals over rows and impressions lie below 1, and the block interval reaches 1 only within the simulation noise of the bootstrap. On All and Women's intervals are wider with longer blocks and contain 1 at every length tried. Also post hoc, with blocks of 5 to 180 minutes the lower end of the pooled change, on which one registered label rests, lies between 0.001 and 0.004 and is within two Monte Carlo standard errors of 0 at three of the six lengths; the half-width behind another label, that of a self-normalised estimate of the uniform policy, lies between 0.093 and 0.102 against a bound of 0.10. Post hoc distances that read no click, fixed after the registered results and before they were computed, put the whole-week frequencies of the BTS log at 0.386 and 0.413 from the prior and the hourly frequencies at 3.0 and 3.4 times the noise floor from the whole-week ones, above the threshold at which the registered rule for that mismatch finds a difference, and without an interval; they bound the registered distance and do not say how much of it is change. One value of each hashed user feature is written differently in the files of the two policies; for one feature per confirmatory campaign that is a post hoc inference. A semi-synthetic study, registered separately after all of these results, keeps the rows of the logs and generates clicks with known dependence: under dependence that reaches one to two minutes, solved from published statistics of the logs, nominal 95% intervals of the IPS ratio from rows or impressions cover 0.85 to 0.88 and those from blocks of five minutes or longer 0.94 to 0.95, and on Women's the ten-minute intervals are not shown to cover.

1 Introduction

Whether an off-policy estimator works on real logs can be checked only where the value of the evaluated policy is also known. The Open Bandit Dataset (Saito et al., 2021a) offers such a case and is used for it (Saito et al., 2021d; Saito and Joachims, 2022; Gupta et al., 2024; Felicioni et al., 2024; Jeunen, 2025a): it holds one week of logs of a uniform-random policy and of a Bernoulli Thompson sampling (BTS) policy from the same site, so the click rate of BTS can be estimated from the random log and compared with the click rate of the BTS log, the on-policy value.

On the Men's campaign the inverse propensity score (IPS) estimate is 0.891 of the on-policy value; the notebook published with Jeunen (2025a) arrives at the same figure, and its bootstrap interval for IPS excludes the on-policy value (Section 2.4). The evaluated distribution is computed from a published Beta prior with no updates, and public documents do not say that this prior was the policy in force during the

logged week: the software describes it as the prior used at the start of data collection, and an open issue reports recorded propensities that do not match it and change within seconds (Section 2.3). The rows of the logs are also not independent, while the published intervals treat them as if they were.

This report presents a pre-registered study of how the estimate changes when the evaluation target is aligned with the log. Propositions, statistics, decision rules and random streams were committed before any outcome of the campaigns All and Women's was computed. Men's served as the development campaign and enters no verdict; the registered rules were applied to All and Women's, the confirmatory campaigns, in one run. The registration is a commit in the author's repository, made 79 minutes before that run, when the outcome on Men's, public point estimates for the other two campaigns and the issue's report on All were known (Section 3.2).

Registered results on the confirmatory campaigns (Section 4).

- The published prior is not the distribution the BTS log shows: the total variation distance between the two, by hour, is 0.526 on All and 0.527 on Women's, against noise floors of 0.136 and 0.115, and the registered label is "inconsistent" on both. This quantifies, with a noise floor and on two campaigns, the mismatch that the issue reported for All.

- No shortfall is shown. The IPS ratio is 0.948 (0.867 to 1.033) on All, 1.004 (0.917 to 1.099) on Women's and 0.974 (0.915 to 1.036) pooled, with intervals from a bootstrap over ten-minute blocks; the self-normalised estimator gives the same labels. These intervals exclude a pooled shortfall above 8.5% and no smaller one.

- With the target aligned to the frequencies of the BTS log in the same three-hour period, the estimate is not distinguishable from the on-policy value in any verdict unit. The paired change from the fixed-prior estimate is 0.072 (0.002 to 0.139) pooled, labelled "raises the estimate", and "no change shown" on All and on Women's; the lower end of the pooled interval is about two Monte Carlo standard errors from 0. Because no shortfall was established, the registered rules do not read the gap, and the primary question has no answer.

- The frequencies of the BTS log in the hour after a row are associated with that row's click in the random log more than those of the hour before. An aligned target is therefore not exogenous to the clicks it is evaluated on; the direction of the resulting bias is not established.

Post hoc analyses (Section 5), descriptive and without verdicts.

- The standard deviation of the ratio over block replicates is 1.31 to 1.47 times the one over rows or impressions on all three campaigns. On Men's the intervals over rows and impressions lie below 1; the block interval of IPS ends at 1.0046, which is 1 within the simulation noise of the bootstrap. On All and Women's the interval contains 1 under each resampling unit.

- With blocks of 5, 10, 20, 30, 60 and 180 minutes, fixed before the analysis was run, the intervals of All, Women's and the pooled unit contain 1 at every length and are wider at every length above ten minutes, so the ten-minute intervals may be too narrow there.

- The intervals behind the labels of the aligned estimate, of the association in time and of the reverse direction were recomputed with the same block lengths; quantities and reading were fixed after the registered results and before the run (Section 5.3). The lower end of the pooled change lies between 0.0011 and 0.0040 and is within two Monte Carlo standard errors of 0 at three of the six lengths. The half-width of the self-normalised estimate of the uniform policy lies between 0.0927 and 0.1015 against a bound of 0.10 and is within two of its errors of the bound at five of the twelve combinations of campaign and length. The interval of the difference behind the association in time lies above 0 with blocks of 5 and of 20 minutes on both campaigns; it contains 0 with blocks of 60 minutes on both and of 180 minutes on Women's, where at 30 minutes its lower end is within two errors of 0.

- Three distances that read no click, fixed after the registered results and before they were computed, look inside the mismatch of the first registered result (Section 5.5). The whole-week frequencies of the BTS log are 0.386 and 0.413 from the prior, and the hourly frequencies are 0.410 and 0.394 from the whole-week ones, 3.0 and 3.4 times the noise floor. By the thresholds of the registered rule, and without an interval, the hourly frequencies thus differ from those of the whole week; the distances bound the registered one and do not say how much of it is change.

- One value of each hashed user feature is written differently in the files of the two policies, on 22.7% to 35.4% of the rows of a random log depending on the campaign. For three of the four features this follows from an amendment made before the confirmatory run, for the fourth from a post hoc inference.

Table 1: Items and rows of the six log files, and the on-policy value V_{on}, the click rate of the BTS log, with its interval from the registered block bootstrap (Section 3.4).

	Items	Rows, random log	Rows, BTS log	V_{on}
All	80	1,374,327	12,357,200	0.004953 (0.004876 to 0.005032)
Women's	46	864,585	7,765,497	0.006379 (0.006271 to 0.006486)
Men's	34	452,949	4,077,727	0.006743 (0.006596 to 0.006893)

A separate study of coverage (summarised in Section 7), registered on its own after all of the above, with generated clicks on the kept rows of the logs.

- Under dependence that reaches one to two minutes, solved from published statistics of the logs, nominal 95% intervals of the IPS ratio cover 0.85 to 0.88 with rows or impressions as resampling units and 0.94 to 0.95 with blocks of five minutes or longer. On Women's the standard deviation of the real logs at 180 minutes over the one at ten exceeds the 97.5th percentile of that scenario, and by the rule of that study the ten-minute intervals there are reported as not shown to cover.

The report does not show why the estimate on Men's is about a tenth below the on-policy value, whether a shortfall below the size the test could detect exists on the other campaigns, or what the deployed parameters were.

2 The benchmark

2.1 Data

The Open Bandit Dataset (Saito et al., 2021a) holds the logs of a seven-day experiment in late November 2019 on a fashion e-commerce site, in three campaigns called All, Men's and Women's. In the words of the dataset paper, each campaign "randomly uses either the Random policy or the Bernoulli TS policy for each user impression", and a policy selects three items that are shown at three positions. A company blog post on the serving system describes the two policies as allocated to the users who visit the top page (Matsutani, 2020). The unit of assignment is thus described in two ways; it would decide which resampling unit is right, if the files identified it (Section 3.4).

A row of a log is one item at one position: a timestamp, the item, the position, a click indicator, the recorded probability that the logging policy showed the item at the position, and four hashed user features. The dataset paper calls a row a user impression. We use *impression* for a reconstructed page view: a maximal run of rows in time order with the same user features, each less than a second after the one before. The files hold no user or session identifier. Table 1 gives the sizes. The released BTS log of All has 189,116 more rows than the 12,168,084 that the dataset paper reports; the other five files match its counts. Every row of the six files lies in the week from 24 to 30 November 2019 (UTC).

2.2 Target and reference value

The data allow two comparisons. The main tables of the dataset paper estimate the click rate of the random policy from the BTS log. This report concerns the other direction: the uniform-random log is the evaluation data, BTS is the policy to evaluate, and an estimate of its click rate is compared with the click rate of the BTS log, which the dataset paper calls the on-policy estimate.

The distribution of the target comes from the accompanying pipeline. Its `BernoulliTS` class, with the option `is_zozotown_prior`, loads one Beta(α, β) per item from a file. Its method `compute_batch_action_dist` samples every item's Beta distribution, places the items with the three largest samples at the three positions, repeats this 100,000 times by default, and returns the resulting frequency of each item at each position as one matrix repeated for every row. No parameter is updated between draws. The dataset paper says in its main text that the data come with "the exact policy implementations used in real production", and its appendix introduces this step with the words "we now replicate BernoulliTS used during the data collection in the ZOZOTOWN production".[1] We compute the same distribution with 1,000,000 draws and call it the fixed-prior target π_0.

[1] Code statements refer to the files `obp/policy/contextfree.py`, `benchmark/ope/benchmark_ope_estimators.py` and `docs/about.rst` of the pipeline's repository, `st-tech/zr-obp` on GitHub, at commit `8cbd5fa`.

2.3 The public record on the deployed policy

The pipeline's code documents `is_zozotown_prior` as selecting the hyperparameters "used at the start of the data collection period". Versions 2 and 3 of the dataset paper carry a footnote saying that Bernoulli TS was pre-trained for over a month before data collection "and the policy well converges to a fixed one" (Saito et al., 2020, 2021b). The version presented at a workshop in August 2021 has the same footnote and draws the consequence: the dataset "fits the standard OPE formulation, that assumes fixed behavior and evaluation policies" (Saito et al., 2021c). Version 5, the one published, has no such sentence, and neither has the supplement in the proceedings; the pipeline's documentation still has it. Version 5 says in a footnote that the recorded probabilities were computed "by Monte Carlo simulations based on the policy parameters [...] used during the data collection process". The blog post says that for bandits without context the parameters are updated by periodically aggregating the logs, that the parameters used in serving are stored by version, and that the recommendation servers check periodically for updated parameter files (Matsutani, 2020); it gives no interval.

An issue opened in the pipeline's repository in October 2026 reports that recorded probabilities of the All campaign do not match the prior: for one item at the first position the prior gives about 0.198 where the first row of the log records 0.104925. It reports values that alternate within seconds and rows that record exactly the uniform probability, and it asks whether the parameters were updated during the logs and under which assumptions the comparison with a frozen prior is valid (voidprove, 2026). It had no reply from the maintainers when this study was registered; its only comment at the time of writing is a note by the author with the results reported here.

On the Men's log the registration counts, without reading a click, what the issue describes for All. 99.96% of the recorded probabilities are multiples of 1/200,000. An item at a position takes about 850 distinct values over the week, and 59% of the changes of value return within seconds to the value before the previous one. Servers that pick up new parameter files at different moments, as the blog post describes, would produce such a pattern; the log does not show whether that is the cause.

None of these sources states the parameters in force at any time, the update interval, or which version served a row.

2.4 Reported results and their intervals

For this direction the dataset paper reports, in an appendix, the root mean squared error of each estimator over 200 bootstrap samples of 300,000 rows drawn with replacement from the random log, and no interval for an estimate. Its example code prints an IPS estimate of 0.004553 for the All campaign, 1.198 times the click rate of the random log, without saying on how many rows. Version 3 had reported the relative error $|\hat{V} - V_{\mathrm{on}}|/V_{\mathrm{on}}$ averaged over bootstrap samples: 0.051, 0.111 and 0.028 for IPS on All, Men's and Women's with the whole logs, and 0.092, 0.195 and 0.040 when the logs are split by time, a setting that in its words "may introduce the distributional change" between the data and the environment that defines the value (Saito et al., 2021b).

Jeunen (2025a) shows the Men's campaign in a figure with 99% intervals; variances are "estimated through the sample covariance over the logged data". The notebook published with that paper computes the target with `is_zozotown_prior` and 1,000,000 draws. It prints an IPS estimate of 0.006011 (0.005612 to 0.006472), with a 95% interval from 100 bootstrap resamples, against an on-policy value of 0.006743; the ratio is 0.891. In that table the upper ends of all six estimators are at most 0.006597, below the on-policy value. The bars of the figure are the estimate plus and minus 2.567 standard errors; by the standard errors the notebook prints, the upper end is 0.006645 for IPS and 0.006771, 0.006751 and 0.006784 for SNIPS, the doubly robust estimator and IPS with a baseline, so these three intervals contain the on-policy value and that of IPS does not (Jeunen, 2025b). The paper adds that results on the other campaigns were similar, with smaller improvements "due to already well-performing base estimators". Gupta et al. (2024) report a mean relative absolute error of 0.128 for IPS on the same campaign.

Saito and Joachims (2022), Saito et al. (2021d) and Felicioni et al. (2024) use the All campaign and compare estimators with each other over bootstrap samples of rows: by squared error relative to that of IPS, by squared error against the on-policy value with both directions mixed, and by measures of estimator selection. A difference between the reference value and the value of the evaluated target enters every such comparison and is not reported separately.

Two public reproductions on the full data came to our notice. One covers all three campaigns and reports lowest relative errors of 6.7%, 9.3% and 2.2% on All, Men's and Women's, averaged over 10 bootstrap resamples of the random log; it notes that rows of one page view are not independent and that its intervals are therefore somewhat too narrow (RollinsE, 2026). The other reports, for All, an IPS estimate of 0.004692 (0.004382 to 0.005022) against a reference of 0.004953 from 200 bootstrap resamples, and notes that "The

reference is itself a logged CTR estimate" (lyftium, 2026). The notebook and both reproductions call the target the production policy.

A short paper at the Web Conference 2026 examines the dataset (Korovaitceva et al., 2026). On synthetic data it finds that the ranking of estimators by squared error changes with the sample size and with the random state of the generator. On the dataset it runs no estimator and makes four observations, each on the All campaign only. The logging "policy changed during the experiment": a histogram shows the recorded propensities of the pair of item and position whose propensities vary most in the BTS log, and the paper notes that the frequency and timing of the updates are not documented. Because that policy varies over time, the BTS log "can not be used for the ground-truth evaluations without introducing an additional bias, which is hard to quantify", and the authors add: "We are not aware of any estimates of this bias in the literature." Pearson chi-square tests on the random log reject uniformity over the pairs of item and position (2186.15 on 239 degrees of freedom), over the items, and over the items at the first position (752.18 and 749.38 on 79). The dataset holds no user identifier, and several users share a feature vector. The paper gives no interval for an estimate on the dataset, resamples nothing, and does not examine the reuse of lists or dependence between clicks.

Two of these observations meet this report. The bias of the second is the first of the components of Section 3.1, a difference between π_0 and the policy that produced the BTS log. The registered study quantifies the mismatch behind it (H1) and asks how much of a shortfall an aligned target removes; under its rules that question has no reading on the confirmatory campaigns (Section 4), so the estimate that the authors find missing is not supplied here. The third concerns the item counts of the random log, not the recorded propensities of the BTS log that H2 examines. The registration reports a statistic of the same kind for the random log of Men's, a chi-square of 11.0 times its degrees of freedom for the item counts of the week, in the same place as the reuse of lists; the three statistics of the paper are 9.1 to 9.5 times theirs. A Pearson test takes the rows as independent draws, which the rows of reused lists are not (Section 5.1), so its rejection does not by itself show unequal propensities; the mean weight of Section 3.3, the registered check of how often the random log shows the items that π_0 favours, is 1.000 (0.989 to 1.011) on All.

None of the reports named in this subsection explains the difference on Men's, and none but the last discusses whether the deployed policy changed during the week.

3 Design

3.1 What a difference can consist of

With uniform logging and known propensities, IPS is unbiased for the value of any fixed target, whatever the dependence between rows. A systematic difference between the IPS estimate for π_0 and the click rate of the BTS log can therefore consist only of the following: a difference between π_0 and the policy that produced the BTS log; a departure of the random log from its stated propensities; a difference between the two logs in users or in their mix over time; an effect of the other items of a list on a click; and effects in the BTS log itself, such as repeated display. The registered propositions map onto this list. H1 compares π_0 with the BTS log, and H4 and H5 replace π_0 by the log's own frequencies; the mean weight of H3 checks the random log; H7 compares the users; an exploratory item concerns the other items of a list; H2 and H6 check the recorded propensities of the BTS log. Nothing in the design addresses repeated display.

We call an IPS or SNIPS ratio below 1 a *shortfall*. The registration uses *gap* for the same difference where it asks what accounts for it.

3.2 Registration and sample roles

The propositions, statistics, decision rules and random streams of this study were committed to its public repository, `https://github.com/sHyanorn/ope-checks`, before any outcome of the campaigns All and Women's was computed (commit `08ab8da`). We call that document the registration. Nosek et al. (2018) define preregistration as committing to analytic steps without advance knowledge of the research outcomes, and discuss the case of data that already exist and the practice of holding part of the data back. This study is of that kind and departs from the definition in ways that this subsection lists.

Men's is the development campaign. Code was written and checked on it, and its results enter no verdict. All and Women's are the confirmatory campaigns. Nothing was computed from their four log files, beyond row counts and checks of the input that read no click, until the code for every proposition had run on Men's and had been committed. The confirmatory results of all propositions come from one run of that commit (`11e9340`). A first attempt at the run was cut off by a restart of the host. The author's record of that event,

added to the repository after the run, says that no statistic had been written or printed; the same command was then run again at the same commit.

Seven amendments were written while the code was run on Men's: six fix how a sentence of the registration is read where the text allowed two readings or left a case open, and one adds a check of the input files. One of the six, amendment 7, changes how H7 treats the values of a user feature, after the feature columns of Men's showed the problem described in Section 4.6. All were committed before any confirmatory outcome (commits `baa4af5` and `1a28064`; Appendix A).

The registration is a commit in a git history, made 79 minutes before the confirmatory run started, on 10 November 2026, and it carries no timestamp from a third party. The push record of the hosting service is the only outside evidence of the order of events.

The registration lists what was known when the rules were written, and much was. On Men's the outcome was known: an IPS ratio of 0.891 (0.815 to 0.966) with impressions as resampling units and 0.891 (0.791 to 0.994) with ten-minute blocks (1,000 replicates of another random stream), and a ratio of 0.917 (0.848 to 0.989) with the whole-week frequencies of the BTS log as target and impressions as resampling units. The structure of the Men's logs and the dependence of their clicks were known (Section 3.4). For All and Women's, public point estimates were known (Section 2.4), and so was the issue's report that recorded probabilities on All do not match the prior. The registration states as its premise that the published prior is not the deployed policy, and it expected "no shortfall shown" as the likely result of H3 in every confirmatory unit. The outcomes of H1 and H3 were thus foreseeable. What the registration fixed in advance is the choice of statistics and thresholds for them and the rules of H4 and H5. Two choices were shaped by Men's outcomes, and the registration says so: the block length of the bootstrap and the thresholds of H1. Three further settings, the coverage threshold and the minimum cell count of H4 and the eligibility bound of H2, were fixed from quantities of Men's that read no click.

3.3 Estimators

A campaign has n_A items and three positions k. R and B are its random and BTS logs; a row holds a timestamp, an item a, a position k, a click c and a recorded propensity p. The on-policy value is the click rate of the BTS log, $V_{\mathrm{on}} = |B|^{-1} \sum_{i \in B} c_i$. The fixed-prior target $\pi_0(a \mid k)$ is the share of 1,000,000 draws in which item a ranks k-th when every item's $\mathrm{Beta}(\alpha, \beta)$ of the published prior file is sampled once per draw. With weights $w_i = n_A \, \pi_0(a_i \mid k_i)$ on the rows of R,

$$\hat{V}_{\mathrm{IPS}} = \frac{1}{|R|} \sum_{i \in R} w_i c_i, \qquad \hat{V}_{\mathrm{SNIPS}} = \frac{\sum_{i \in R} w_i c_i}{\sum_{i \in R} w_i}. \tag{1}$$

Both treat the pair of item and position as the action, as the benchmark does, and take the propensity of the random log to be $1/n_A$ as recorded. Every estimate is reported as a ratio to V_{on}. The ratio of the two estimates is the mean weight $|R|^{-1} \sum_i w_i$. It has expectation 1 if the random log shows every item equally often at every position, so a mean weight away from 1 says that the random log showed the items that π_0 favours more or less often than its stated propensity implies.

3.4 Bootstrap

The resampling units are the 1,008 ten-minute UTC clock intervals of the week. A replicate draws 1,008 blocks with replacement, and a block carries the rows of every log that fall in it: R and B are resampled together, and so are All and Women's in a pooled quantity. Everything computed from the logs is recomputed in each replicate, V_{on} included. Fixed on the full data are π_0, the period width of H4, the eligible cells of H2 and the impressions of the exploratory item. Intervals are the 2.5th and 97.5th percentiles of 10,000 replicates, and a difference between two estimates is computed within each replicate. The block multiplicities of replicate i are one multinomial draw from a seeded generator, the same for every campaign; Appendix E gives the streams.

This is a non-overlapping block bootstrap: one "resamples disjoint blocks of fixed integer length" with replacement (Nordman, 2009). Equivalently it is a bootstrap over clusters (Cameron and Miller, 2015), with clock intervals as clusters, and it assumes that rows in different ten-minute intervals are independent. The theory in Nordman (2009) and Cameron and Miller (2015) concerns smooth functions of means of a stationary series and standard errors of regression coefficients. It does not cover a click rate that changes over the day, blocks of unequal content, or percentile intervals, so the intervals here are not shown to have their nominal coverage (Section 7). A semi-synthetic study of their coverage, registered separately after the results of this report were known, is summarised there.

Table 2: The registered propositions. "Interval" is the bootstrap interval of Section 3.4. A cell is an item at a position in H2 and an item at a position in a period in H4. H7 and an exploratory item on the other two items of an impression have no rule (Appendix A).

	Statistic	Rule
H1	D: distance between π_0 and the hourly frequencies of B; noise floor D_0	"inconsistent" if $D > 0.10$ and $D > 2D_0$; "not distinguishable from variation within the hour" if $D \le D_0$; else "undetermined"
H2	M: mean over eligible cells of $S(a,k)$, defined below from the recorded propensities	"consistent" if $M \in [0.98, 1.02]$ and at least $\lceil 0.9n \rceil$ of the n cells have 1 in their interval; "inconsistent" if $M \notin [0.95, 1.05]$ and its interval misses $[0.98, 1.02]$; else, or with fewer than five cells, "undetermined"
H3	Ratio of IPS and of SNIPS to V_{on}	"low" if the interval lies below 1; "high" if above; else "no shortfall shown"
H4	$\rho = V_{\text{M}}/V_{\text{on}}$ with the target aligned to B by period; $d = \rho$ minus the IPS ratio	Alignment: "raises" or "lowers the estimate" if the interval of d excludes 0, else "no change shown". Level: "below" or "above the on-policy value" if the interval of ρ excludes 1, else "not distinguishable". The gap is read only where H3 found IPS "low"
H5	IPS with the hourly frequencies of B taken from the hour after a row minus the same from the hour before	"frequencies after a row are associated with clicks in the random log" if the interval lies above 0; "the reverse" if below; else "not shown". Descriptive
H6	IPS and SNIPS of the uniform policy from B with its recorded propensities, over the click rate of R	"off" if the interval excludes 1; "consistent" if it contains 1 with a half-width of at most 0.10; else "undetermined"
H7	Distances between the user features of R and B; change in ρ when rows of R are weighted to the user mix of B	None

Rows of these logs are not independent, which is why rows are not the resampling unit. Served lists are reused: on Men's, 37% of consecutive impressions of the random log and 31% of the BTS log show the same three items in the same order. Clicks cluster: on Men's the variance of the difference in click rate between adjacent blocks of ten minutes or more is 2.0 to 2.5 times what independent impressions would give. Where the unit of randomisation is a user, the user is the unit to resample (Bakshy and Eckles, 2013; Deng et al., 2018); the files hold no user identifier, and the unit of assignment is itself described in two ways (Section 2). On Men's the standard deviation of the IPS ratio was 0.052 to 0.054 for blocks of 5 to 60 minutes and 0.038 with impressions, and the length of ten minutes was chosen after seeing these numbers. No rule for choosing a block length was applied.

The intervals do not include the Monte Carlo error of π_0 or dependence that reaches beyond a block. In the other direction, resampling blocks treats differences between blocks as noise; the joint resampling of both logs removes a daily cycle that is common to them from a ratio, and what does not cancel widens the interval.

3.5 Propositions and decision rules

Table 2 lists the seven propositions; H1 to H4 are the main ones and H4 is primary. A *verdict unit* is a campaign or the pooled unit. Verdict units are All and Women's for every proposition and, for H3 and H4, a pooled unit: $0.54 \times$ (All) $+ 0.46 \times$ (Women's), applied to ratios and to their differences within each replicate. The weights are the shares of expected clicks in the two random logs, computed from the row counts and click rates that the dataset paper reports. Pooled labels are primary and are always shown with those of each campaign. A pooled label is flagged "campaigns differ" if the interval of All minus Women's excludes 0.

H1, the prior against the log. With $N_B(\cdot)$ the row counts of B, h the clock hour and $f_h(a \mid k) = N_B(h,a,k)/N_B(h,k)$,

$$D = \sum_{h,k} \frac{N_B(h,k)}{N_B} \, \mathrm{TV}\big(f_h(\cdot \mid k),\, \pi_0(\cdot \mid k)\big), \tag{2}$$

where TV is half the L_1 distance over the items. The noise floor D_0 is the same sum with π_0 replaced by the frequencies of the first, third and fifth ten-minute blocks of the hour and f_h by those of the second, fourth and sixth. Whole blocks are split because lists are reused: a random split of rows would put copies of one list

in both halves. Under a fixed policy equal to π_0, D is about half of D_0, since D_0 compares two half samples and D one full sample with a fixed distribution. D_0 also contains any real change of the policy within the hour, which makes "inconsistent" harder to reach. H1 reads no click. It compares π_0 with hourly frequencies and so does not separate a fixed policy other than π_0 from a policy that changed from hour to hour.

H2, the recorded propensities. If a row's recorded propensity is the probability with which its item was chosen at its position, then $S(a,k) = N_B(k)^{-1} \sum_{i \in B:\, a_i = a,\, k_i = k} 1/p_i$ has expectation 1 for every item and position. A cell is eligible if it has a row and no row with $p < 0.001$, so that no single row dominates. H2 reads no click. The registration names two biases of S: a cell stays eligible when a parameter version with a smaller propensity was served and never chosen, which biases S downward, and the recorded propensities lie on a grid of 1/200,000, which biases it upward.

H4, aligning the target. Periods are runs of W clock hours, $W \in \{1, 3, 6, 24\}$. With $c_R(b,a,k)$ the click rate of the rows of R that show item a at position k in period b,

$$V_{\mathrm{M}} = \sum_{b,a,k} \frac{N_B(b,a,k)}{N_B}\, c_R(b,a,k). \tag{3}$$

V_{M} uses no propensity: it weights the random log's click rates by how often the BTS log showed each item at each position in each period. A cell is covered if R has at least five rows in it. The width used is the smallest W at which covered cells hold at least 99% of the rows of B, chosen per campaign on the full data from counts alone. A cell that is not covered takes c_R from the narrowest wider period that has five rows, up to the whole week; a cell without five rows in the week is dropped, and a campaign in which dropped cells hold more than 0.1% of the rows of B gets no H4 label. In a replicate the width stays and counts, coverage and fallback are recomputed. The paired change d compares two estimates that differ in two respects, the target and the form of the estimator: V_{M} does not depend on how often the random log shows each item, and IPS does. The registration therefore also reports the aligned target in IPS form and IPS with the whole-week frequencies of B as target, without a rule.

Under the rule of H4 the gap is read only in a verdict unit where H3 called IPS "low"; elsewhere the reading is "shortfall not established; gap not read", whatever the two labels of H4 are. Since the registration expected H3 not to find a shortfall, it expected the gap to go unread; what it could deliver in any case are the labels of d and ρ.

When the aligned estimate is biased. The registration says that V_{M} estimates the value of the deployed policy only if the BTS shares within a period do not depend on the random log's clicks in that period, and that if BTS learned from clicks of both logs' users, or follows demand that both share, V_{M} is too high, more so at small W. A short calculation, made after the results, supports the first half of that sentence and not the second. Let $\pi_t(a)$ be the share with which BTS shows an item at time t within a period and $q_t(a)$ the click probability, the same for the users of both logs, with equal traffic profiles. The on-policy value of the period is the time average of $\sum_a \pi_t(a) q_t(a)$, while V_{M} converges to $\sum_a \bar{\pi}(a)\, \bar{q}(a)$, which is that average minus $\sum_a \mathrm{Cov}_t(\pi_t(a), q_t(a))$. If BTS follows demand that both logs share, the covariance within a period is positive and V_{M} is too low, less so as W shrinks. An upward bias arises if the BTS shares depend on the noise in the random log's own clicks, as they would if the parameters were updated from the clicks of both logs. H5 is the registered, descriptive check of an association in time; it is positive under both mechanisms and cannot tell them apart. The direction of the bias of V_{M} is therefore not established, and ρ is reported at every width.

4 Registered results

Every number in this section comes from the one confirmatory run, except in Section 4.7, which gathers the development campaign. Table 3 gives the statistics and labels of the confirmatory verdict units; Appendix B holds further quantities that the registration reports without a rule, and the result files hold all of them. No pooled label carries the flag "campaigns differ", and no quantity was undefined in any replicate.

4.1 H1: the published prior is not the distribution the BTS log shows

On both campaigns D exceeds 0.10 and twice its noise floor, and the label is "inconsistent": D is 0.526 against a D_0 of 0.136 on All and 0.527 against 0.115 on Women's (Figure 1).

The registration also applies the statistic to a case where the policy is known, the random log against the uniform distribution. There D is 0.186 and 0.172 on All and Women's, which is 0.53 and 0.53 of its noise

Table 3: Registered results on the confirmatory campaigns: estimate and interval (2.5th to 97.5th percentile of 10,000 block replicates) and the label the registered rule gives. The pooled unit exists for H3 and H4 only. ρ and d are at the period width the rule selected, three hours on both campaigns.

	All	Women's	Pooled
H1 distance D; noise floor D_0	0.526; 0.136	0.527; 0.115	
H1 label	inconsistent	inconsistent	
H3 IPS ratio	0.948 (0.867 to 1.033)	1.004 (0.917 to 1.099)	0.974 (0.915 to 1.036)
H3 SNIPS ratio	0.948 (0.867 to 1.032)	1.004 (0.917 to 1.099)	0.974 (0.914 to 1.036)
H3 label, both estimators	no shortfall shown	no shortfall shown	no shortfall shown
H4 aligned estimate ρ	1.031 (0.932 to 1.128)	1.062 (0.974 to 1.158)	1.045 (0.978 to 1.114)
H4 paired change d	0.083 (−0.019 to 0.182)	0.058 (−0.031 to 0.147)	0.072 (0.002 to 0.139)
H4 alignment label	no change shown	no change shown	raises the estimate
H4 level label	not distinguishable	not distinguishable	not distinguishable
H4 reading	gap not read	gap not read	gap not read
H5 hour after minus hour before	0.099 (0.033 to 0.166)	0.076 (0.022 to 0.130)	
H2 eligible cells; mean M	3; 0.973 (0.957 to 0.990)	7; 0.967 (0.955 to 0.980)	
H2 label	undetermined	undetermined	
H6 IPS ratio, uniform policy	0.990 (0.918 to 1.070)	0.956 (0.897 to 1.019)	
H6 SNIPS ratio, uniform policy	0.950 (0.855 to 1.050)	0.920 (0.819 to 1.014)	
H6 label, both estimators	consistent	consistent	

floor, close to the half that a fixed policy is expected to give. For the BTS log against the prior the same quotient is 3.9 and 4.6.

The pipeline describes the prior as the one used at the start of data collection (Section 2.3), so the registration reports the distance in the first hour of the week: 0.180, 0.126 and 0.088 by position on All and 0.164, 0.098 and 0.105 on Women's. No noise floor was registered for a single hour, and the registered results do not say whether these numbers show a mismatch at the start. As Section 3.5 notes, the label does not say either whether the log shows another fixed policy or one that changed. A post hoc analysis takes up both points with distances that read no click (Section 5.5): the hourly frequencies differ from those of the whole week by 3.0 and 3.4 times the noise floor; of the six first-hour distances, three are at most a floor computed for that hour and three lie between the floor and twice the floor. That comparison rests on one hour and one split, and we do not read it as showing or excluding a mismatch at the start.

4.2 H3: no shortfall is shown

The IPS ratio is 0.948 (0.867 to 1.033) on All, 1.004 (0.917 to 1.099) on Women's and 0.974 (0.915 to 1.036) pooled. The SNIPS ratios agree with them to the third decimal except in two interval ends (Table 3). Every interval contains 1, so the label is "no shortfall shown" for both estimators in all three verdict units (Figure 2). The mean weight is 1.000 (0.989 to 1.011) on All and 1.000 (0.989 to 1.012) on Women's: by this check the random logs of both campaigns show the items that π_0 favours as often as their stated propensity implies.

The label is not a finding of no shortfall. The pooled interval excludes ratios below 0.915, that is, shortfalls of more than 8.5%, and no smaller one. Section 5.2 shows post hoc that this bound depends on the block length.

4.3 H4: the aligned estimate, and a gap that the rules do not read

One-hour periods cover 0.9712 and 0.9775 of the rows of the BTS log and three-hour periods 0.9985 and 0.9992, so the rule selects three hours on both campaigns; 0.15% and 0.08% of the rows are resolved by fallback and none is dropped.

The aligned estimate ρ is 1.031 (0.932 to 1.128) on All, 1.062 (0.974 to 1.158) on Women's and 1.045 (0.978 to 1.114) pooled, and the level label is "not distinguishable from the on-policy value" in every verdict unit. The paired change d from the fixed-prior IPS ratio is 0.083 (−0.019 to 0.182), 0.058 (−0.031 to 0.147) and 0.072 (0.002 to 0.139). The alignment label is "no change shown" on each campaign and "raises the estimate" in the pooled unit. The pooled label sits at its threshold: the lower end of its interval, 0.002, is 2.1 Monte Carlo standard errors of such a percentile from 0 (about 0.0009 under a normal approximation; this arithmetic is post hoc). Post hoc, with blocks of 5 to 180 minutes that lower end lies between 0.0011 and 0.0040 and is within two Monte Carlo standard errors of 0 at 20, 30 and 60 minutes (Section 5.3). The point

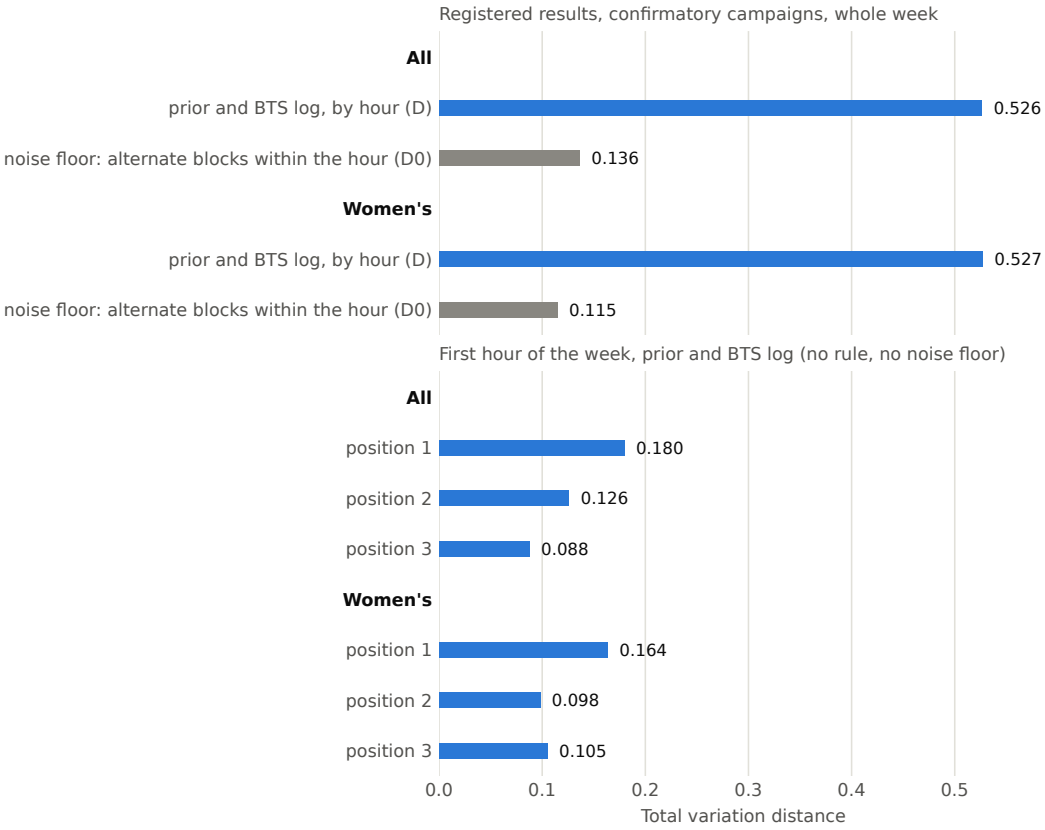


Figure 1: H1 on the confirmatory campaigns (registered). Top: distance D between the published prior and the hourly frequencies of the BTS log, and its noise floor D_0 from alternate ten-minute blocks within the hour. Bottom: the distance in the first hour of the week by position, which enters no rule and has no registered noise floor.

Table 4: The path from the fixed prior to the aligned target in steps, as ratios to the on-policy value (registered quantities; only the last column and its difference from the first carry a rule). The second column is IPS with the BTS log's item frequencies of the whole week as target, a fixed target; the third is IPS with its frequencies by three-hour period; the fourth is ρ, which also changes the form of the estimator.

	Fixed prior, IPS	Whole week, IPS	Three hours, IPS	Three hours, ρ
All	0.948 (0.867 to 1.033)	1.017 (0.949 to 1.087)	1.047 (0.948 to 1.152)	1.031 (0.932 to 1.128)
Women's	1.004 (0.917 to 1.099)	1.006 (0.939 to 1.077)	1.093 (1.001 to 1.194)	1.062 (0.974 to 1.158)
Men's (development)	0.891 (0.790 to 1.005)	0.917 (0.822 to 1.017)	0.976 (0.851 to 1.100)	0.992 (0.853 to 1.114)

estimate of d is positive at each of the four widths on both campaigns (Table 10); these eight estimates come from nested periods of the same rows, and every one of their intervals contains 0.

Because H3 found no shortfall, the reading in every verdict unit is "shortfall not established; gap not read". The study's primary question therefore has no registered answer on the confirmatory campaigns.

Table 4 shows the steps that the registration reports without a rule. Replacing the prior by the whole-week frequencies of the BTS log, a fixed target for which IPS on a uniform log is unbiased, moves the point estimate from 0.948 to 1.017 on All and from 1.004 to 1.006 on Women's. Making the target local in time moves it further, to 1.047 and 1.093; on Women's the interval of that estimate, 1.001 to 1.194, lies above 1. ρ is 1.083 with one-hour periods and 1.018 with whole days on Women's, and 1.016 and 1.017 on All (Table 10). No rule applies to these steps, and none of the differences between columns has an interval. As far as point estimates go, on Women's almost all of the movement comes from the step that makes the target local in time, which is the step whose estimate may be biased (Section 3.5), and on All most of it comes from the fixed whole-week target.

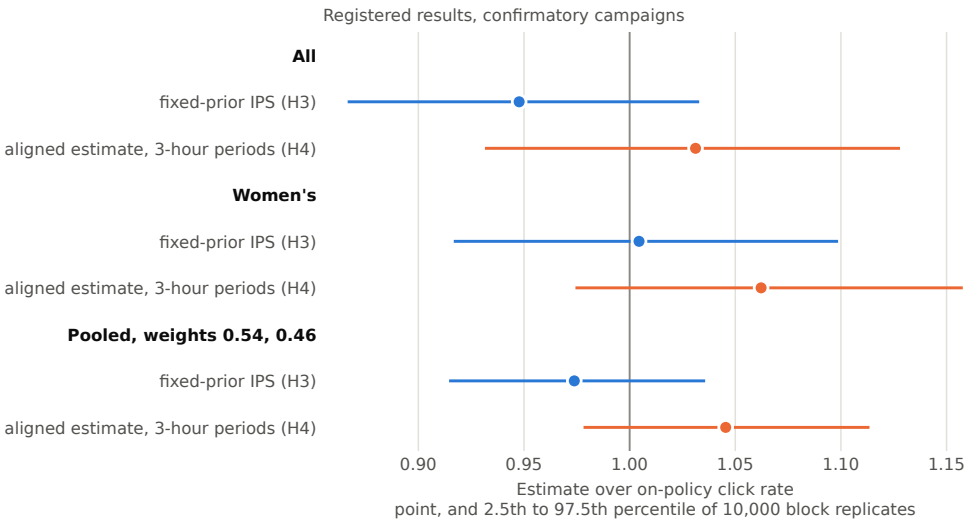


Figure 2: Registered estimates on the confirmatory campaigns as ratios to the on-policy value: the fixed-prior IPS estimate of H3 and the aligned estimate ρ of H4, with their block-bootstrap intervals.

4.4 H5: frequencies after a row are associated with clicks in the random log

H5 builds the target from the BTS frequencies of three windows around each row of the random log. With the hour before the row the estimate is 1.002 of the on-policy value on All and 1.036 on Women's; with the hour after it is 1.101 and 1.124. The paired difference is 0.099 (0.033 to 0.166) and 0.076 (0.022 to 0.130), and the label on both campaigns is the one in the heading (Table 11). Post hoc, the interval of the difference also lies above 0 with blocks of 5 and of 20 minutes on both campaigns; it contains 0 with blocks of 60 minutes on both campaigns and of 180 minutes on Women's, where at 30 minutes its lower end is within two Monte Carlo standard errors of 0 (Section 5.3).

What BTS shows after a row is thus associated with what the random log's users clicked before. The registration calls H5 descriptive because it cannot separate BTS learning from the clicks of the random log's users from demand that both logs share. Under the first mechanism ρ is biased upward; under the second it is not (Section 3.5). H5 is compatible with a biased ρ and does not show one. It also uses one-hour windows in IPS form, not the three-hour ρ.

4.5 H2 and H6: two checks of the recorded propensities

H2 is "undetermined" on both campaigns, so nothing is claimed about the recorded propensities in either direction. All has 3 eligible cells, fewer than the five the rule needs; their M is 0.973 (0.957 to 0.990). Women's has 7 cells with an M of 0.967 (0.955 to 0.980), which lies outside $[0.98, 1.02]$ and inside $[0.95, 1.05]$, and 4 of the seven cell intervals contain 1. M is below 1 on both campaigns, in the direction of the downward bias the registration names.

H6 estimates the uniform policy from the BTS log with its recorded propensities; it is the direction of the dataset paper's main tables. Both estimators are "consistent" with the random log's click rate on both campaigns: IPS is 0.990 (0.918 to 1.070) on All and 0.956 (0.897 to 1.019) on Women's. For SNIPS the label sits at its threshold: the half-widths are 0.097 and 0.098 against the bound of 0.10. Post hoc, with blocks of 5 to 180 minutes the half-width is 0.0927 to 0.1015 and within two of its Monte Carlo standard errors of 0.10 at five of the twelve combinations of campaign and length (Section 5.3). Kish's effective sample size is 19,910 of 12,357,200 rows on All and 6,009 of 7,765,497 on Women's, and rows with a recorded propensity below 0.001 carry 39% and 29% of the weight.

4.6 H7 and the exploratory item

H7 compares the user features of the two logs. On Men's the feature columns showed, during development and without reading a click, that each of the four hashed features has exactly one value that occurs only in the random log and one that occurs only in the BTS log, on the same rows in all four features: 29.21% of the random log and 29.35% of the BTS log. Compared as written, the distance of every feature is 0.29; with the two values taken as one it is 0.001 to 0.003. Amendment 7, made before any confirmatory outcome was

Table 5: *Post hoc.* Intervals of the H3 ratios under three resampling units, and the standard deviation (SD) of the block replicates over that of the row and impression replicates. The block column reproduces the registered intervals.

			Interval			SD of blocks over	
		Estimate	Rows	Impressions	Blocks	rows	impressions
Men's	IPS	0.891	0.818 to 0.964	0.816 to 0.968	0.790 to 1.005	1.47	1.44
	SNIPS	0.908	0.834 to 0.983	0.832 to 0.987	0.805 to 1.022	1.47	1.43
All	IPS	0.948	0.888 to 1.009	0.885 to 1.011	0.867 to 1.033	1.37	1.31
	SNIPS	0.948	0.888 to 1.008	0.885 to 1.011	0.867 to 1.032	1.36	1.31
Women's	IPS	1.004	0.942 to 1.068	0.941 to 1.071	0.917 to 1.099	1.45	1.40
	SNIPS	1.004	0.942 to 1.067	0.941 to 1.070	0.917 to 1.099	1.45	1.41

computed, therefore treats two values of a feature as one if exactly one value occurs only in each log, and compares values as written in every other case.

On each confirmatory campaign that condition held for three of the four features, whose distances are 0.001 to 0.002 on All and 0.001 to 0.003 on Women's. It failed for feature 2 on All and feature 3 on Women's, which were compared as written. The registered joint distance of 0.360 and 0.236, the share of BTS rows whose combination of features is absent from the random log, 0.353 and 0.228, and the change in ρ when rows are weighted to the user mix of the BTS log, -0.058 (-0.136 to 0.013) and -0.116 (-0.178 to -0.063), therefore still contain a difference in how the two files write one value. They cannot be read as a difference between the users of the two logs (Section 5.6).

The exploratory item asks whether a click on an item depends on the two other items shown with it: -0.013 (-0.092 to 0.063) on All and 0.050 (-0.033 to 0.132) on Women's, as a share of the click rate.

4.7 The development campaign

Men's enters no verdict. Its registered run gives a D of 0.437 against a D_0 of 0.109, an IPS ratio of 0.891 (0.790 to 1.005) and a SNIPS ratio of 0.908 (0.805 to 1.022). The registration quotes an interval of 0.791 to 0.994 from 1,000 replicates of another stream; the registered stream with 10,000 replicates puts the upper end above 1. The mean weight is 0.981 (0.968 to 0.994), with an interval below 1: the random log of Men's showed the items that π_0 favours less often than uniform draws would, which is the whole difference between the IPS and SNIPS ratios there. The registration also records, for the item counts of that log against the uniform distribution, a chi-square statistic over its degrees of freedom of 6.9 across hours and 11.0 over the week. The path of Table 4 is 0.891, 0.917, 0.976 and 0.992, with a d of 0.100 (-0.010 to 0.190). M of H2 is 1.005 (0.997 to 1.013) over 12 cells, and the SNIPS label of H6 is "undetermined".

The three IPS ratios, 0.891, 0.948 and 1.004, are not shown to differ: the registration defines no test of a difference between campaigns other than All minus Women's, whose interval contains 0, and we make none. The study started from the Men's number, the largest of the three deviations; a number that draws attention by its size tends to overstate what is behind it.

5 Post hoc analyses

Everything in this section was fixed after the confirmatory results were known. The analyses are descriptive, give no verdict and replace no registered label. Wherever ten-minute blocks are resampled, the multiplicities are the registered ones.

5.1 The resampling unit

The registration measured on Men's alone how much narrower an interval over impressions is. Table 5 gives the two ratios of H3 an interval under three resampling units on every campaign: rows, impressions and the registered ten-minute blocks. The comparison changes two things at once: rows and impressions are drawn in each log on its own, and blocks jointly in both logs.

With blocks the standard deviation is 1.31 to 1.47 times the one with rows or impressions, on all three campaigns and for both estimators. On All and Women's every interval contains 1 under each unit, so the registered label of H3 on the confirmatory campaigns does not rest on the choice of blocks. On Men's the intervals from rows and from impressions lie below 1 (Figure 3). The block interval of SNIPS ends at 1.022.

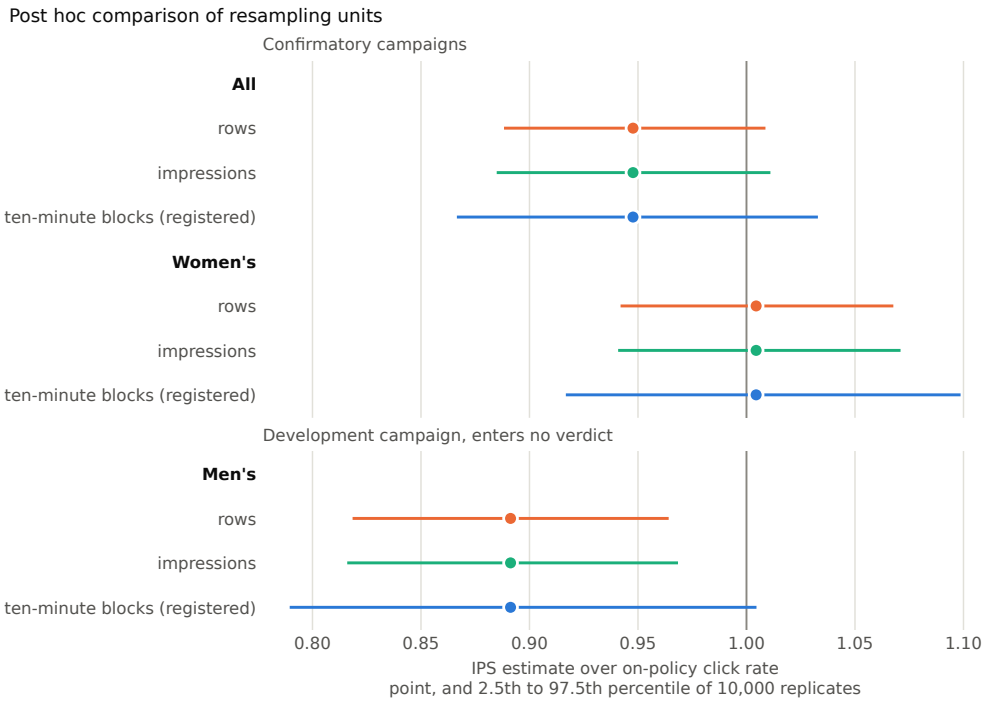


Figure 3: *Post hoc.* The IPS ratio of H3 with its interval under three resampling units.

That of IPS ends at 1.0046, which Section 5.2 and Section 5.8 show to be 1 within the simulation noise of the bootstrap: on Men's, resampling blocks leaves undecided what resampling rows or impressions decides.

The registration reports two measures of dependence for Men's, and Appendix C gives them for every log. An impression is complete if it has exactly three rows that hold positions 1, 2 and 3 once each. In each of the six logs 30% to 37% of complete impressions show the same three items in the same order as the complete impression before them. Under independent uniform draws that share would be 0.0002% in the random log of All, 0.0028% in that of Men's and 0.0011% in that of Women's. This measure reads no click. The second does: the variance of the difference in click rate between adjacent ten-minute blocks is 1.85 to 2.45 times the variance that difference would have if impressions were independent. The ratio carries no interval and does not separate dependence between impressions from a click rate that drifts from one block to the next.

One figure of the registration did not survive this analysis. It gives a variance ratio of 1.2 to 1.5 for one-minute blocks on Men's. That value lacks a small-sample factor in the variance of a block's click rate, without which the ratio exceeds 1 for independent impressions in blocks that hold few of them. With the factor the one-minute ratio on Men's is 1.18 and 1.20. The figure entered no rule, and the factor changes the ratios at ten minutes by at most 0.04.

5.2 The block length

The registration says that the width of the intervals does not depend on the block length between 5 and 60 minutes, and that this was measured on the fixed-prior ratios of Men's only. For this analysis the lengths (5, 10, 20, 30, 60 and 180 minutes), the random stream and the reading were committed before it was run (commit `72df47d`; the run is of commit `7ec1253`). A block of L minutes is a UTC clock interval counted from the start of the week; a replicate draws as many blocks as the week holds, jointly in both logs and on every campaign. The reading fixed beforehand was: state where each interval contains 1 and the range of the standard deviation relative to ten minutes; treat lengths whose standard deviations differ by under 2% as not told apart, given the Monte Carlo error of 10,000 replicates; and take a standard deviation that grows with the length as pointing to dependence between rows, or to change over the week, that reaches beyond ten minutes, without separating the two.

On All, Women's and the pooled unit the interval of each ratio contains 1 at every length (Table 6).

On All, Women's and the pooled unit the standard deviation is smaller at 5 minutes than at ten and larger at every longer length: 0.95 to 0.97 of the ten-minute value at 5 minutes, 1.07 to 1.12 at 60 minutes and 1.13 to 1.36 at 180 minutes, across units and estimators. On Men's the differences from ten minutes go in both directions and reach 2.9%; 4 of the 10 combinations of length and estimator exceed the 2% of the reading. Differences of 2% to 3% therefore occur on a campaign that shows no trend, and the smallest

Table 6: *Post hoc.* Interval of the IPS ratio of H3 with clock blocks of several lengths as resampling units, and the standard deviation of the replicates over the one at ten minutes. The ten-minute column is the registered interval. The week holds 2,016 blocks of 5 minutes and 56 of 180. Table 13 has the SNIPS ratio and the standard deviations.

		5 min	10 min	20 min	30 min	60 min	180 min
Men's	interval	0.790 to 1.001	0.790 to 1.005	0.790 to 1.001	0.789 to 1.000	0.788 to 1.004	0.781 to 1.004
	SD ratio	0.98	1.00	0.98	0.98	1.01	1.03
All	interval	0.870 to 1.028	0.867 to 1.033	0.866 to 1.033	0.863 to 1.037	0.863 to 1.040	0.856 to 1.044
	SD ratio	0.95	1.00	1.02	1.05	1.07	1.13
Women's	interval	0.919 to 1.095	0.917 to 1.099	0.910 to 1.104	0.908 to 1.108	0.906 to 1.112	0.890 to 1.144
	SD ratio	0.96	1.00	1.05	1.09	1.12	1.36
Pooled	interval	0.916 to 1.035	0.915 to 1.036	0.911 to 1.039	0.907 to 1.042	0.909 to 1.042	0.897 to 1.056
	SD ratio	0.97	1.00	1.04	1.09	1.10	1.31

difference on the confirmatory units above ten minutes, 2.1% (All, IPS, 20 minutes), is of that size. Whether the rise on the confirmatory units exceeds sampling error is not tested: a standard deviation estimated from n blocks has an error of its own, roughly $1/\sqrt{2(n-1)}$ of its value under a normal approximation, which is 5.5% at 168 blocks and 9.5% at 56, and the estimates at different lengths come from the same rows. That error figure was not part of the reading fixed beforehand.

If the rise reflects dependence beyond ten minutes, the ten-minute intervals of the confirmatory campaigns are too narrow; if it reflects change over the week that does not cancel in the ratio, the longer blocks overstate the width (Section 3.4). The analysis does not decide between the two. *Coverage study.* A registered extension of the semi-synthetic study did not separate them either (Section 7). In the separately registered semi-synthetic study (Section 7), which is not one of the post hoc analyses of this section, the standard deviation at 180 minutes over the one at ten lies between 0.80 and 1.22 (All) and between 0.80 and 1.24 (Women's) from the 2.5th to the 97.5th percentile of simulated data sets whose dependence reaches only one to two minutes; the published 1.13 lies inside that range and the published 1.36 above it. The pooled IPS interval is 0.915 to 1.036 at ten minutes, 0.909 to 1.042 at 60 minutes and 0.897 to 1.056 at 180 minutes, so the shortfall that the interval excludes moves from more than 8.5% to more than 10.3%. This analysis covers the two ratios of H3; Section 5.3 does the same for the intervals behind the other labels.

On Men's the upper end of the IPS interval lies between 0.9997 and 1.0046 across the six lengths and falls below 1 at 30 minutes only.

5.3 The block length and the intervals of H4, H5 and H6

Two registered labels rest on intervals that end close to the value their rule compares them with: the pooled alignment label of H4, whose interval of d has its lower end at 0.002, and the SNIPS label of H6, whose half-widths are 0.097 and 0.098 against the bound of 0.10 (Sections 4.3 and 4.5). For this analysis the quantities and the reading were committed before it was run (commit `02eb3da`; the run is of commit `c5eb6c5`). The quantities are ρ and d of H4 at three hours, the difference of H5 between the hour after and the hour before, and the two ratios of H6. Blocks, replicates and multiplicities are those of Section 5.2; the width of H4 stays as chosen on the full data, and counts, coverage, fallback and windows are recomputed in every replicate. The pooled unit is formed for ρ and d only, as registered.

The reading fixed beforehand has four parts. The registered rules are defined for ten-minute blocks, so at another length we say where an interval lies and give no label. An end of a percentile interval of 10,000 replicates carries a Monte Carlo standard error of about 0.027 standard deviations of the replicates, and a half-width about 0.019, under a normal approximation; an end less than two such errors from the value it is compared with, or a half-width less than two of its errors from 0.10, is reported as at that value and counted on neither side. A standard deviation that rises with the length is read as in Section 5.2. And 12 intervals per length are compared with a value on the confirmatory units, without allowance for their number.

The interval of ρ contains 1 in every unit at every length, and the interval of d contains 0 on All and on Women's at every length (Table 14). In the pooled unit the lower end of d lies between 0.0011 and 0.0040 at the six lengths (Table 7). By the reading it lies above 0 at 5, 10 and 180 minutes and at 0 at 20, 30 and 60 minutes. At no length does the interval reach below 0. On Men's the interval of d contains 0 at every length but 5 minutes, where its lower end, -0.0025, is at 0.

The SNIPS interval of H6 contains 1 at every length on both campaigns, and its half-width is 0.0927 to 0.1015. By the reading the half-width is below 0.10 at 5, 10 and 180 minutes on both campaigns and at 20 minutes on All, and at 0.10 at 30 and 60 minutes on All and at 20, 30 and 60 minutes on Women's; at 20 and 30 minutes on Women's the number itself is above 0.10. The IPS interval of H6 contains 1 at every

Table 7: *Post hoc.* The intervals that end close to the value they are compared with, by block length. The ten-minute column holds the registered intervals. After a semicolon: the distance of the lower end of d from 0, and of the half-width of the SNIPS interval of H6 from 0.10, in Monte Carlo standard errors, counted positive on the side of the registered label. Table 14 has every quantity and unit.

		5 min	10 min	20 min	30 min	60 min	180 min
Pooled	d	0.003 to 0.137	0.002 to 0.139	0.001 to 0.144	0.001 to 0.150	0.001 to 0.147	0.004 to 0.132
	lower end; errors	0.0031; 3.3	0.0020; 2.1	0.0014; 1.5	0.0015; 1.5	0.0011; 1.1	0.0040; 4.6
All	H5 difference	0.035 to 0.165	0.033 to 0.166	0.025 to 0.171	0.012 to 0.184	−0.022 to 0.220	0.023 to 0.169
Women's	H5 difference	0.025 to 0.129	0.022 to 0.130	0.015 to 0.138	−0.001 to 0.153	−0.039 to 0.196	−0.005 to 0.140
All	SNIPS half-width; errors	0.0927; 8.3	0.0974; 2.8	0.0977; 2.5	0.0995; 0.6	0.0985; 1.6	0.0938; 7.0
Women's	SNIPS half-width; errors	0.0957; 4.7	0.0976; 2.6	0.1015; −1.6	0.1004; −0.4	0.0998; 0.2	0.0959; 4.5

length with a half-width of 0.059 to 0.081. On Men's, where the registered SNIPS label is "undetermined", the half-width is 0.178 to 0.189.

For the two labels that sit at their thresholds at ten minutes the reading thus gives: the lower end of the pooled d lies above 0 at three of the six lengths and at 0 at three, and the SNIPS half-width lies below 0.10 at seven of the twelve combinations of campaign and length and at 0.10 at five. In no case does an end lie on the other side of its value by that reading.

The H5 difference lies above 0 at 5, 10 and 20 minutes on both campaigns. On All it also does at 30 and 180 minutes, and its interval at 60 minutes, −0.022 to 0.220, contains 0. On Women's the lower end is at 0 at 30 minutes, and the intervals at 60 and 180 minutes, −0.039 to 0.196 and −0.005 to 0.140, contain 0. The standard deviation of the difference is largest at 60 minutes, 1.79 and 2.14 times the one at ten minutes, and smaller again at 180 minutes (1.08 and 1.33). On Men's the interval lies above 0 at every length, 60 minutes included (0.010 to 0.304).

The reading did not foresee that a block length can interfere with the construction of an estimate, and the following remark was not fixed beforehand. H5 takes its target from the clock hour before and the clock hour after a row and uses only rows that have both. A clock hour is missing from a replicate with probability 0.0024 at ten minutes and with probability 0.05, 0.13 and 0.37 at 20, 30 and 60 minutes; with 60-minute blocks a replicate holds both neighbours of a given hour with probability 0.40. At 180 minutes a three-hour block is missing with probability 0.36, and two of the three hours of a block have a neighbour in another block. These figures follow from the multinomial draw and from no data. The H5 intervals at 20 minutes and more therefore rest on fewer rows than those at ten. How many rows the replicates leave out was not recorded, so the part of the widening that this accounts for is not known, and we read the widening neither as a sign of dependence nor as explained by this remark. Where the interval of the H5 difference would lie with longer blocks and all rows kept is not answered by this analysis.

For ρ the standard deviation relative to ten minutes is 1.00 to 1.13 in the pooled unit, 0.97 to 1.11 on All and 0.96 to 1.24 on Women's; for d it is 0.95 to 1.09, 0.84 to 1.10 and 0.98 to 1.14. At 180 minutes a block is one three-hour period of H4. The standard deviation does not rise with the length throughout: at 180 minutes it is below the ten-minute value for d on All and in the pooled unit and for ρ on All. As in Section 5.2, differences between lengths are not attributed.

5.4 What the confirmatory rule could detect

The rule of H3 calls a ratio "low" when the upper end of its interval is below 1. With a replicate standard deviation s and a normal approximation, a true ratio of $1 - \delta$ is called "low" with probability $\Phi(\delta/s - 1.96)$: one half at $\delta = 1.96\,s$ and 0.8 at $\delta = 2.80\,s$. This arithmetic was not registered. It assumes that s does not change with δ, that the replicate standard deviation is the true sampling standard deviation, which Section 5.2 gives reason to doubt, and that the interval is symmetric about the estimate. The standard deviations are those of Table 13.

For the pooled IPS ratio at ten minutes the two values of δ are 6.1% and 8.7%; with the standard deviation at 180 minutes they are 8.0% and 11.4%. A single campaign needs more: 11.9% on All and 13.1% on Women's for a probability of 0.8 at ten minutes. A shortfall of 10.9%, the point estimate on the development campaign and as such a selected one (Section 4.7), would have been called "low" in the pooled unit with a probability of 0.94 at ten minutes and 0.76 with the standard deviation at 180 minutes; one of half that size, 5.4%, with 0.42 and 0.27.

5.5 The prior, the whole week and the hours

The label of H1 does not say whether the BTS log shows one fixed distribution that is not the prior's or a distribution that changed during the week (Section 3.5). Three distances that read no click bear

Table 8: *Post hoc.* Distances that read no click. D and D_0 are the registered statistic of H1 and its noise floor; A sets the target against the whole-week frequencies of the log and E the hourly frequencies against the whole-week ones. The target is the prior for the BTS log and the uniform distribution for the random log. The reference of A is the same distance between alternate ten-minute blocks.

	Log	D	D_0	A	Reference	E	$D-A$	E/D_0	$(D-A)/D_0$
Men's	BTS	0.437	0.109	0.310	0.010	0.335	0.127	3.06	1.16
	random	0.202	0.366	0.018	0.029	0.202	0.185	0.55	0.50
All	BTS	0.526	0.136	0.386	0.012	0.410	0.141	3.01	1.03
	random	0.186	0.354	0.016	0.028	0.185	0.170	0.52	0.48
Women's	BTS	0.527	0.115	0.413	0.010	0.394	0.115	3.44	1.00
	random	0.172	0.325	0.013	0.027	0.172	0.160	0.53	0.49

on this. They and their reading were committed before they were computed (commit `02eb3da`). With $f(a \mid k) = N_B(a,k)/N_B(k)$ the frequencies of the whole week,

$$A = \sum_k \frac{N_B(k)}{N_B}\,\mathrm{TV}\big(f(\cdot \mid k),\, \pi_0(\cdot \mid k)\big), \qquad E = \sum_{h,k} \frac{N_B(h,k)}{N_B}\,\mathrm{TV}\big(f_h(\cdot \mid k),\, f(\cdot \mid k)\big), \tag{4}$$

so that A sets the prior against the whole week and E the hours against the whole week. The third is a noise floor for the first hour of the week: the distance, by position, between the frequencies of the first, third and fifth ten-minute block of that hour and those of its other blocks. A is shown with a reference, the same distance between the whole-week frequencies of alternate ten-minute blocks. All of this is also computed for the random log with the uniform distribution in place of π_0, a case in which the policy is fixed and known. No interval is computed, as for D and D_0 in the registration.

Because the whole-week frequencies are the row-weighted mean of the hourly ones and TV is convex, $A \le D$; by the triangle inequality, $D \le A + E$. If the log showed one fixed distribution all week, E would hold noise only and $D - A$ would be at most that noise. The noise reference is D_0: if the policy is fixed within an hour and the two halves of the hour are independent, the expected distance between the hour's frequencies and the distribution of its policy is at most the expected distance between the halves, and about half of it when counts are large. The reading fixed beforehand sets E and $D - A$ against the two thresholds of the registered rule of H1: at most D_0, where that rule says "not distinguishable from variation within the hour"; above $2D_0$ and above 0.10, where it finds a difference; and between the two, where nothing is said. The first-hour distances are set against their floor and twice their floor in the same way. The reading also says what the distances cannot do: total variation distances do not add, so A and E bound D from both sides and do not divide it into a part due to a fixed difference and a part due to change.

On the BTS logs of All and Women's A is 0.386 and 0.413, against references of 0.012 and 0.010 (Table 8), that is 32 and 40 times the reference; the reading fixed no threshold for A. E is 0.410 and 0.394, which is 3.0 and 3.4 times D_0 and above the upper threshold on both campaigns; by the reading this says that the distribution the BTS log shows changed between hours. In the random log, whose policy did not change, E is 0.52 to 0.55 of D_0 on the three campaigns, and A is below its reference.

$D - A$ is 0.141 on All and 0.115 on Women's. On both it lies between D_0 and $2D_0$ (on Women's it exceeds D_0 by 0.0001), so by the reading nothing is said on whether the change between hours adds to the distance from the prior beyond what the whole week's frequencies already hold. A and E are 0.73 and 0.78 of D on All and 0.78 and 0.75 on Women's, and they do not add to it. The registered label of H1 stands. The distances add, without an interval, that the hourly frequencies lie above the upper threshold from those of the whole week and that A is many times its reference; how much of D is change is not determined. On Men's the pattern is the same: A is 0.310, E is 0.335, 3.1 times D_0, and $D - A$ lies between the thresholds.

In the first hour of the week the distance from the prior is at most its floor at 3 of the 6 positions of the two confirmatory campaigns and between the floor and twice the floor at the others (Table 15); at the first position it lies between them on both campaigns, at 1.30 and 1.42 times the floor. In the random log the first-hour distance from the uniform distribution is 0.38 to 0.74 of its floor. By the thresholds of the registered rule the first hour is thus not shown to differ from the prior, and three of its six distances lie where the reading says nothing. Not fixed beforehand is the further observation that every first-hour distance, taken by position, is smaller than A, which is an average over positions. None of this is a finding that the prior was in force in that hour: it is one hour with one split, and its floor contains whatever change the policy made within the hour.

The distances describe what the log shows. A distribution that changes between hours can come from updated parameters or from servers that hold different versions of them (Section 2.3), and change within an hour is part of D_0, not of E. Earlier versions of the dataset paper say that the policy "well converges to a

Table 9: *Post hoc.* A separate registered check on the Men's campaign, recomputed with ten-minute blocks. First column: the estimate with its block interval, both divided by the on-policy value as a fixed number, as that project's rule uses them. Second and third columns: under other random streams, how often that interval contains the on-policy value, and the range of its upper end over the on-policy value. DM is the direct method and DR the doubly robust estimator.

	Estimate and block interval	Streams that contain it	Upper end across streams
DM	0.881 (0.875 to 0.888)	0 of 100	0.888 to 0.889
IPS	0.8909 (0.7894 to 1.0015)	67 of 100	0.996 to 1.006
SNIPS	0.908 (0.806 to 1.020)	100 of 100	1.014 to 1.024
DR	0.897 (0.798 to 1.004)	98 of 100	0.999 to 1.009

fixed one" (Section 2.3); in the released week the hourly frequencies differ from those of the whole week by more than the registered rule of H1 asks for a difference.

5.6 The user feature the amendment left unmatched

Amendment 7 left one feature unmatched on each confirmatory campaign (Section 4.6). In that feature, by the feature columns alone, the BTS log holds two values that the random log lacks, and the random log one value that the BTS log lacks. One of the two BTS values sits on 30 rows on All and 14 on Women's. The other, and the random log's value, sit on exactly the rows that the matched values of the other three features mark: 485,993 rows of the random log and 4,362,623 of the BTS log on All, 195,980 and 1,771,203 on Women's.

Taking those two values as one, the registered code recomputes H7 on a random log in which the value is written as the BTS log writes it. The joint distance between the user features of the two logs falls from 0.360 to 0.007 on All and from 0.236 to 0.009 on Women's, the level of the registered reference, the same distance between alternate blocks of one log (0.006 and 0.014 on All, 0.006 and 0.015 on Women's). The change in ρ under user weights becomes 0.001 (-0.003 to 0.005) and -0.001 (-0.014 to 0.005), against a registered -0.058 (-0.136 to 0.013) and -0.116 (-0.178 to -0.063); on Men's, where the amendment matched all four features, the registered change is -0.001 (-0.007 to 0.004). Every quantity outside H7 comes out as registered. That the two values are one category is an inference from the rows they sit on; the files do not state it (Table 16).

For anyone who joins the two logs on user features, one consequence does not depend on that inference. Between 22.7% and 35.4% of the rows of a random log, by campaign, carry in each feature a value that no row of the BTS log of that campaign carries, and 22.8% to 35.3% of the rows of a BTS log carry one that the random log lacks.

5.7 Where the propensity statistic falls short

M of H2 is 0.027 below 1 on All and 0.033 on Women's. Call an hour low for an eligible cell if the cell is shown in it at under a tenth of the rate that the propensities of its recorded rows imply. Low hours account for 0.033 of the distance of S from 1 on All and 0.022 on Women's, averaged over the eligible cells, against 0.002 on Men's, where M is 1.005; the figure can exceed the distance of M from 1 because other hours lie above 1. At the registered eligibility bound 0.031 of the 0.033 on All and 0.021 of the 0.022 on Women's fall on 27 and 29 November. At lower bounds the pattern by day is weaker (Appendix D). Such hours are what the downward bias named in the registration would leave; rows missing from the release would leave them too. The logs do not show what was served in those hours, and the cause is not identified.

5.8 An application: a registered check on the development campaign under block resampling

A separate project of the author, push-lab (`https://github.com/sHyanorn/push-lab`), had registered a check of four estimators on the full Men's campaign: an estimator passes if the interval of its estimate contains the on-policy value, taken as a fixed number, and its median relative error is below 0.5. With impressions as resampling units no estimator passed. After the results above, that project recomputed its intervals post hoc (its decision record D063, commit `937ad77`) with the ten-minute blocks and the random stream of this study, leaving target, reward model and point estimates unchanged (Table 9). Its interval differs from the ratio interval of this report in holding the on-policy value fixed, and its target was computed with other Monte Carlo draws.

With blocks the intervals of IPS, SNIPS and DR contain the on-policy value and that of DM does not. For IPS this turns on the random stream: under 100 other streams its interval contains the on-policy value in 67, that of DR in 98, that of SNIPS in all and that of DM in none. That project kept its registered verdict and calls no estimator validated. For the question of Section 5.1 it shows that the upper end of a block interval for IPS on Men's moves across 1 with the seed.

6 Implications for users of the benchmark

The evaluated distribution is not what the BTS log shows. A difference between an off-policy estimate for the fixed-prior target and the click rate of the BTS log mixes the error of the estimator with the difference between that target and whatever produced the log (H1, Section 4). Such an offset does not cancel when estimators are compared with each other against the same reference: it favours the estimator whose error happens to point towards it. Post hoc, by the thresholds of the registered rule of H1 and without an interval, the log does not show one distribution either: the hourly frequencies differ from those of the whole week by 3.0 and 3.4 times the noise floor (Section 5.5). A fixed target can then at most match the frequencies of the week. As one of the public reproductions puts it, "The reference is itself a logged CTR estimate" (lyftium, 2026).

Resample time, jointly in both logs. Rows and impressions are not independent in these logs, and the standard deviation over block replicates is 1.31 to 1.47 times the one over rows or impressions (post hoc). The on-policy value is itself an estimate from a log with the same dependence and belongs inside the resampling. Ten minutes is not a calibrated block length. *Coverage study.* On generated clicks with dependence that reaches one to two minutes, nominal 95% intervals of the IPS ratio from rows or impressions cover 0.85 to 0.88 and those from blocks of five minutes or longer 0.94 to 0.95; with added dependence across an hour ten-minute blocks cover 0.91 on All and 0.85 on Women's; that study does not show which structure the real logs have (Section 7). If the wider intervals at longer blocks on All and Women's (post hoc) reflect dependence, the advice of Cameron and Miller (2015) to "use bigger and more aggregate clusters when possible, up to and including the point at which there is concern about having too few clusters" points to blocks of about an hour, of which the week holds 168; if they reflect change over the week, longer blocks overstate the uncertainty. Reporting more than one length shows the reader how much depends on it: here two registered labels rest on interval ends that are within simulation error of their thresholds at some lengths (post hoc, Section 5.3). A remark made after those results, whose size was not measured: with blocks as long as the hour windows of H5 the replicates leave out rows that the estimate needs.

Report all three campaigns. The estimate about a tenth below the on-policy value is that of Men's. On All and Women's the registered IPS ratio is 0.948 and 1.004 with intervals that contain 1, also under the other resampling units and block lengths tried (post hoc). The three ratios are not shown to differ (Section 4.7).

Aligning the target to the log is not shown to be a remedy. Replacing the prior by the BTS log's own frequencies in the same period changes the estimate by 0.072 (0.002 to 0.139) in the pooled unit, with the label "no change shown" on each campaign. The aligned target is not exogenous to the clicks it is evaluated on (H5), and the direction of the bias this causes is not established (Section 3.5). A target made of the log's frequencies over the whole week is fixed, and IPS on a uniform log is unbiased for it; the registered estimates for that target are in Table 4.

Match the hashed user features before joining the two logs. One value of each user feature is written differently in the random file and in the BTS file: by the registered amendment for all four features on Men's and for three on All and on Women's, and by a post hoc inference for the fourth there. Compared as written, the joint distribution of the user features of the two logs is 0.360 and 0.236 apart in total variation on All and Women's; with the values matched it is 0.007 and 0.009 (post hoc). A model fitted on the contexts of one file and applied to the other meets a value it has not seen on 22.7% to 35.4% of the rows, depending on the campaign (Section 5.6). We know of no published work that joins the two logs on user features.

The deployed parameters could be recovered only from the parameter versions that the serving system is said to have stored (Matsutani, 2020).

7 Limitations

In order of the threat to the conclusions.

The intervals are not validated on the real logs. The block bootstrap was chosen with development results in view, and on the real logs its coverage cannot be checked against a known truth. Time blocks are clusters of unequal size, and at 180 minutes there are 56 of them; Cameron and Miller (2015) report that inference with few clusters rejects too often, "few" being by their account anywhere from under 20 to under 50 clusters, that tests "can over-reject considerably if the clusters are unbalanced", and that resampling whole clusters does not cure the first problem. Dependence that reaches beyond three hours, such as visits of one user spread over days, is not covered by any length tried. The statement that intervals over rows are too narrow rests on the structure of the logs, on a comparison of standard deviations and on the generated clicks of the study described next, not on a demonstration of correct coverage for blocks on the real logs. The conclusion that needs no narrow interval is that on All and Women's no shortfall is shown under any resampling unit or block length tried (post hoc for all but ten-minute blocks).

A separate semi-synthetic study of coverage. This paragraph reports a study of its own, kept in the same repository (`obd_coverage/`). Its click model, scenarios, number of data sets and reading were committed after every result of this report was known and before any of its code (commit `f507c89`), and its parameters before any click was generated (commit `90f9a44`); the run is of commit `df8019a`. None of its statements is a registered or a post hoc result of the study reported in Sections 4 and 5, and it changes no label there. It keeps the rows of the logs and generates their clicks with known dependence, so that the value of every quantity is known, on 4,000 data sets per scenario and campaign with intervals of 2,000 replicates. It calls a coverage adequate if the Wilson interval of the simulated coverage lies within 0.925 to 0.975 and under-covering if that interval lies below 0.925. With dependence within impressions and within synthetic sessions of one to two minutes, solved from the published ratio of standard deviations between impressions and rows and the published variance ratios at ten minutes, nominal 95% intervals of the IPS ratio cover 0.87 (All) and 0.85 (Women's) with rows as resampling units and 0.88 and 0.86 with impressions, all under-covering, and 0.94 to 0.95 with blocks of 5 to 180 minutes, all adequate. That scenario gives the whole of the published variance ratio to dependence; if part of the ratio is a click rate that moves between blocks, rows and impressions cover more than this. With added dependence across an hour, solved from the widening of the intervals at 180 minutes, ten-minute blocks cover 0.91 and 0.85; with dependence across a day every block length under-covers, at 0.82 to 0.89. On Women's the published standard deviation at 180 minutes over the one at ten, 1.365, exceeds the 97.5th percentile of that ratio over the data sets of the short-range scenario, 1.239; by the rule that study registered, the ten-minute intervals on Women's are then reported as not shown to cover. On All the published ratio, 1.134, lies below that percentile, 1.217. The nesting of the click model caps a strength at that of the next narrower grouping, and on All and Women's the short-range scenario on Women's, the hour scenario on Women's and the day scenario on both were cut back in this way; they are weaker than the published numbers they were solved from. No scenario reproduces every published statistic of a campaign. A separately registered extension of that study (commit `af7e615`) measured the dependence of the real clicks directly and added a drift in one log; it changes no label above. The three results above on All and Women's, that rows and impressions under-cover and every block length is adequate under dependence of one to two minutes and that ten-minute blocks under-cover with dependence across an hour, kept their labels when the session size, the size of the long groups and the shape of the group factor were varied one at a time. In a scenario fitted to the excess of double clicks within an impression and to the variance ratios at one and ten minutes, measured on the real clicks, blocks of 5 to 180 minutes are adequate; rows and impressions cover 0.90 and 0.91 on All, under-covering, and 0.92 and 0.93 on Women's, unresolved, with Wilson intervals that contain 0.925; the registration of the extension counts this against the first of the three results there. That scenario reproduces 16 of 25 statistics of the logs on All and 11 on Women's, each within the 2.5th to 97.5th percentile over its data sets, and on no campaign the published ratio of the standard deviation over ten-minute blocks to the one over rows, which lies above that range: its intervals widen less from rows to blocks than those of the logs. Two scenarios built on it match the published standard deviation at 180 minutes over the one at ten on Women's: with dependence across an hour ten-minute blocks cover 0.83, under-covering, and with a drift in the BTS log alone 0.96, adequate. The registered observable meant to tell these apart, the share of the block variance of the ratio at 180 minutes that comes from pairs of rows of different user types or logs, lies on All and on Women's inside both of its registered ranges, the one from the scenarios without a drift in one log and the one from the drift in the BTS log, so neither campaign is separated. On Women's, where it is 0.31, this label rests on the scenario with dependence across a day, which was cut back; without that scenario the label would have been

more like a drift. The study and its extension do not show which structure the logs have. The pooled unit and the intervals behind H4, H5 and H6 were not simulated.

The main result is a null result, and two labels sit at their thresholds. "No shortfall shown" leaves a pooled shortfall of up to 8.5% unexcluded at ten minutes and, post hoc, up to 10.3% at 180 minutes. The pooled alignment label of H4 and the SNIPS label of H6 rest on a lower end that stays between 0.0011 and 0.0040 and on a half-width that stays between 0.0927 and 0.1015, against 0.10, at every block length tried (post hoc, Section 5.3). At some lengths each is within simulation error of its threshold, so the side on which it falls can depend on the block length and on the random stream. Post hoc, the interval of the H5 difference lies above 0 with blocks of 5 and of 20 minutes and contains 0 with blocks of 60 minutes on both campaigns and of 180 minutes on Women's. With long blocks the replicates leave out rows that the estimate needs, a remark made after these results; how much of the widening it accounts for was not measured, and where the interval would lie with longer blocks and all rows kept is open.

The confirmatory campaigns are not independent of the development campaign. All three campaigns come from one site and one week. The registration says that whether the 80 items of All are the 34 of Men's and the 46 of Women's cannot be told from the data. The prior file tells more (a check made after the results, without reading a click, and recorded in the study's repository): all 34 pairs of Beta parameters of Men's occur, identical and in the same order, among the 80 pairs of All, which are all distinct, and none of the pairs of Women's does. One of the public reproductions notes that All "shares many items with" the other two campaigns (RollinsE, 2026). All is therefore not a clean hold-out from Men's, on which the rules were developed; Women's shares no parameter pair with Men's.

What the reference value and the aligned estimate measure. The click rate of the BTS log is the average outcome of a policy whose recorded propensities changed during the week and whose hourly frequencies differ from those of the whole week by more than the registered rule of H1 asks for a difference (post hoc, Section 5.5); it is not the value of any fixed policy that the public record lets us write down. That analysis bounds the registered distance of H1 from both sides and does not say how much of it is change. The released BTS log of All holds 189,116 rows more than the dataset paper reports, 1.5% of the reference sample, and the registration lists what they are among the things the data cannot show. Li et al. (2011) observe that a user who is shown the same article repeatedly is less likely to click, and count "distinct views" to remove this; with 30% to 37% of consecutive impressions repeating a list, that effect may be present in both logs here and is not examined. The aligned estimate V_{M} may be biased, in a direction that is not established (Section 3.5). The effect of dependence between the three positions of an impression cannot be attributed with these data.

One dataset, two estimators. On the confirmatory campaigns only IPS and SNIPS were run. The direct method and the doubly robust estimator appear only in the application of Section 5.8, on the development campaign. The evaluated policy is a learning algorithm that the benchmark freezes; replaying it on the random log, as Li et al. (2011) do for learning algorithms and Dudík et al. (2012) for policies that depend on the history, would give a second estimate and was not done.

The form of the registration. It is a commit in the author's own repository, 79 minutes before the confirmatory run, without a third-party timestamp, written after the development outcomes and public numbers for the other campaigns were known (Section 3.2). The outcomes of H1 and H3 were foreseeable.

The post hoc analyses. Three analyses had their settings and reading fixed before they were run: the block length for H3, the block length for the other intervals, and the distances of Section 5.5. All three were fixed after the confirmatory results were known, and the last two after the earlier post hoc results as well. The reading of the second did not foresee that long blocks interfere with the estimate of H5; that is reported as found and marked as not fixed beforehand. The daily breakdown of H2 computed 63 intervals without allowance for their number, and its measure of low hours replaced an earlier one after a first run.

8 Related work

Section 2.4 covers the work that uses this benchmark. Here we place the checks and the interval method.

Estimators and checks of logged propensities. Strehl et al. (2010) evaluate a policy from logged data by weighting each logged reward with the inverse of a propensity; they require the logging policies to be fixed in advance and do not cover logging that depends on the history. Dudík et al. (2011) add the doubly robust estimator and allow the logging policy to depend on the history. Swaminathan and Joachims (2015) normalise by the sum of the importance weights, which they use as a control variate with known expectation. Lefortier et al. (2016) turn that expectation into a check of the logged propensities of a public advertising dataset, find that for a uniformly random target the mean weight falls well below 1 when several slots are filled, and restrict their benchmark to one slot. The mean weight reported with H3 is this check applied to the random log, and it fails on Men's (Section 4.7); H6 is a check of the same kind on the BTS log. Li et al. (2014) describe two tests of logged propensities, which they call the arithmetic and the harmonic mean test. The summand of H2, an indicator of the shown action over its propensity, is the first term of the identity behind their harmonic mean test; their arithmetic mean test needs the propensity of an action on rows where it was not shown, which these logs do not record. Hadad et al. (2021) show that with data collected by a bandit algorithm the inverse-propensity estimate of an arm's value, with the true adaptive propensities, is unbiased but need not be asymptotically normal. For the value of a policy estimated from adaptively collected data, Zhan et al. (2021) find that the doubly robust estimate stays unbiased but can have high variance and heavy tails, and Bibaut et al. (2021) that neither it nor the inverse-propensity estimate is in general asymptotically normal, so that standard intervals "fail to provide correct coverage". Both reweight the terms of the estimate and require logging probabilities that do not decay too fast. Kato and Kaneko (2020) treat a logging policy that keeps its probabilities fixed within a batch and updates them between batches, with a fixed number of batches of growing size, and apply their estimator to a log of the design of H6: Thompson sampling, updated every 30 minutes, beside uniformly random sampling. Waudby-Smith et al. (2024) give confidence sequences that allow a logging policy that depends on the history and a policy value that changes over time. Each of the four takes the true logging probabilities as given in its main result. H6 uses the adaptively collected BTS log as logging data with recorded propensities that are Monte Carlo estimates and whose check, H2, is undetermined; its interval is a percentile interval over time blocks, which none of these analyses, in the parts read, covers. Jeunen (2025c) derives the probability that an action's draw is the largest under Thompson sampling in closed form, for Beta posteriors with integer parameters; the probabilities of the second and third position, which this dataset records as well, are not derived there.

Comparing offline estimates with online values. Li et al. (2011) validate offline replay on uniformly random logs against a concurrent serving bucket. Their evaluated policy is not given to them either: "We extracted the serving policy" from the serving bucket as the best article in every five-minute interval, which is a precedent for the aligned target of H4. They also list repeat visits of a user among the ways in which logged events fail to be independent. Dudík et al. (2012) evaluate policies that depend on the history by combining rejection sampling on the log with a doubly robust estimate; they assume a reward distribution that "does not change over time" and note that the reward cannot be estimated accurately "from a single trace". Bottou et al. (2013) and Li et al. (2014) compare counterfactual estimates with a second policy run concurrently, and Gilotte et al. (2018) compare estimators against 39 online tests.

Uncertainty with dependent observations. Bakshy and Eckles (2013) run tests without a true effect on click data of users and items and find that a bootstrap which treats observations as independent rejects far too often, nearly 50% of the time in one two-week experiment. They leave experimenters the choice between resampling users, which tests a sharp null correctly, and a multiway bootstrap over users and items, "a somewhat conservative measurement of uncertainty". Deng et al. (2018) treat the case in which the unit of randomisation is coarser than the unit of analysis and show that a variance computed as if analysis units were independent underestimates the true one. For regression, Cameron and Miller (2015) review the same downward bias of default standard errors under correlation within clusters and note that "larger and fewer clusters have less bias but more variability". All three rely on an identifier of the cluster. This dataset has none for users, so we cluster on time (Section 3.4). Kim (2026) examines the first position of the random log of All and reports that the relative click rates of items change over the seven days, with the caveat that "sparse clicks limit its interpretation"; change of that kind is what a block bootstrap counts as noise. Jagerman et al. (2019) study IPS when the reward distribution changes over time under fixed logging and target policies: the estimate over the whole log is then biased for the value at the current time, with a bound that grows with time. The reference of this benchmark is an average over the week, about which their result says nothing. Wu et al. (2025) model A/B tests whose outcome means depend on the time of day and the day of the week; even with a constant assignment probability the standard estimators then "can have larger than optimal variance", and the authors stratify on time. Their outcomes are independent given the time, which is not established for these logs.

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A Registered rules not spelled out in Section 3, and the amendments

Reading of the gap under H4. Only in a unit where H3 called IPS "low". If alignment raises the estimate and the level is not distinguishable: the gap shrinks, and the residual cannot be told from zero and is at most one minus the lower end of the interval of ρ; never "fully explained". If alignment raises it and the level is below: the gap shrinks and a residual remains. If alignment shows no change or lowers and the level is below: alignment is not shown to reduce the gap. If the level is above: the aligned estimate exceeds the on-policy value. If alignment shows no change or lowers and the level is not distinguishable: cannot be determined.

H5. The target of H4 in IPS form at one hour is $\sum_{h,k}[N_B(h,k)/N_B]$ times the mean over the rows of R in (h,k) of $n_A\, g(a \mid k)\, c$. The frequencies g of B at position k are taken from the fifteen-minute clock interval before the one that contains the row, from the clock hour before the row's hour, or from the clock hour after it. The difference of the last two uses only rows that have both windows; by amendment 2 both estimates are recomputed on those rows before they are subtracted.

H6. From B with recorded propensities p and weights $w = 1/(n_A p)$: IPS, the mean of wc, and SNIPS, $\sum wc / \sum w$, of the uniform policy, as ratios to the click rate of R. A row with a very large weight enters IPS only if it was clicked, so "off" on the low side can occur with correct propensities and is not read as evidence against them.

H7. Total variation distance between R and B of each user feature, of their joint distribution and of the hourly shares of rows, each shown with the same distance between the odd and the even ten-minute blocks of one log. Then H4 with each row of R weighted by the ratio of the shares of its combination of user features in B and in R, reported as the change in ρ.

Exploratory item. In R, within each item and position, the click rate of rows whose two companion items have a prior mean above the cell's median minus that of the other rows, averaged over cells and expressed as a share of the click rate.

Amendments. (1) A replicate in which a quantity is undefined is left out of its interval and counted; a quantity behind a label that is undefined in more than 1% of replicates gets no label. (2) The difference of H5, as above. (3) An interval contains a value if the value lies between its ends, ends included. (4) The tolerance for calling a recorded propensity off the grid. (5) Details of the exploratory item. (6) The run also stops if a row has a position other than 1, 2 or 3. (7) The matching of user-feature values described in Section 4.6.

B Registered results in full

The same aligned target in IPS form at the width used gives 1.047 (0.948 to 1.152) on All and 1.093 (1.001 to 1.194) on Women's, and IPS with the whole-week frequencies of the BTS log as target gives 1.017 (0.949 to 1.087) and 1.006 (0.939 to 1.077).

For H6 the SNIPS ratios are 0.950 (0.855 to 1.050) on All and 0.920 (0.819 to 1.014) on Women's. For H2, 48 recorded propensities on All and 2,778 on Women's are off the grid of 1/200,000, and the mean of $1/p$ by position is 79.48, 86.84, 83.60 on All (80 items) and 47.91, 51.07, 44.51 on Women's (46 items).

C Post hoc tables

The tables of this appendix belong to Sections 5.1, 5.2, 5.3, 5.5 and 5.6.

Table 10: H4 at every period width, each with the fallback applied from that width: coverage $C(W)$, ρ, and the paired change from the IPS and from the SNIPS ratio of H3. Registered, reported without a rule except at the width used (3 h).

	W	$C(W)$	ρ	ρ minus IPS ratio	ρ minus SNIPS ratio
All	1 h	0.9712	1.016 (0.911 to 1.106)	0.068 (−0.046 to 0.163)	0.068 (−0.045 to 0.163)
	3 h	0.9985	1.031 (0.932 to 1.128)	0.083 (−0.019 to 0.182)	0.084 (−0.019 to 0.182)
	6 h	1.0000	1.028 (0.935 to 1.117)	0.080 (−0.016 to 0.168)	0.080 (−0.014 to 0.167)
	24 h	1.0000	1.017 (0.944 to 1.091)	0.069 (−0.006 to 0.142)	0.069 (−0.004 to 0.141)
Women's	1 h	0.9775	1.083 (0.980 to 1.190)	0.079 (−0.028 to 0.178)	0.079 (−0.027 to 0.179)
	3 h	0.9992	1.062 (0.974 to 1.158)	0.058 (−0.031 to 0.147)	0.058 (−0.029 to 0.147)
	6 h	1.0000	1.058 (0.972 to 1.147)	0.053 (−0.030 to 0.133)	0.053 (−0.028 to 0.134)
	24 h	1.0000	1.018 (0.945 to 1.093)	0.013 (−0.057 to 0.081)	0.014 (−0.055 to 0.079)

Table 11: H5: the target built from the BTS frequencies of three windows around each row, as ratios to the on-policy value, and the paired difference of the last two (registered).

	Quarter hour before	Hour before	Hour after	After minus before
All	1.004 (0.887 to 1.142)	1.002 (0.902 to 1.116)	1.101 (1.000 to 1.212)	0.099 (0.033 to 0.166)
Women's	1.068 (0.954 to 1.204)	1.036 (0.944 to 1.131)	1.124 (1.028 to 1.225)	0.076 (0.022 to 0.130)

D Post hoc: where the propensity statistic of H2 falls short

The upward bias from the grid of recorded propensities is about 0.0002 at the registered bound, by a first-order calculation. The hourly S of a cell is its sum of $1/p$ in an hour over the rows of its position in that hour, and an hour is low for the cell if the hourly S is under 0.1. The measure replaced an earlier one, hours without any row of a cell, after a first post hoc run. Low hours account for 0.033 of the distance of S from 1 on All and 0.022 on Women's, averaged over the eligible cells, against 0.002 on Men's, where M is 1.005. Low hours are picked by the value of the statistic itself, so some shortfall is found in any log; Men's shows how much on a campaign whose M is not below 1.

By day, M is below 1 with an interval that excludes 1 on 27 and 29 November on both confirmatory campaigns and on no day on Men's (Table 18), and 0.031 of the 0.033 on All and 0.021 of the 0.022 on Women's fall on those two days. The days were looked at after the results were known, and 63 daily intervals were computed without allowance for their number. The pattern by day holds at the registered bound; at the lower bounds of Table 17 other days stand out on All, as the study's post hoc notes record.

E Random streams and reproduction

The block multiplicities of replicate i, from 0 to 9,999, are the same for every campaign, and element j belongs to the j-th ten-minute block of the week:

```
numpy.random.default_rng([20261110, 0, i]).multinomial(1008, numpy.full(1008, 1/1008))
```

The prior draws of campaign c (All 0, Men's 1, Women's 2) are

```
numpy.random.default_rng([20261110, 1, c]).beta(alpha, beta, size=(1000000, n_A))
```

Rows and impressions in Section 5.1 use the streams `[20261110, 2, c, u, l]` with unit u and log l, and blocks of $L \neq 10$ minutes in Section 5.2 use `[20261110, 3, L, i]`. The confirmatory run used numpy 2.5.3.

The repository `https://github.com/sHyanorn/ope-checks` holds the registration (`obd/PREREGISTRATION.md`, commit `08ab8da`), the amendments (`obd/AMENDMENTS.md`), the code and every result file, with the commands that rerun them: `python -m obd.run all women` for the confirmatory campaigns and `python -m obd.posthoc.run` for the post hoc analyses. The confirmatory results are the output of commit `11e9340`, stored in commit `d6a1594`; the post hoc results of Sections 5.1, 5.6 and 5.7 are the output of commit `41969f3`, those of Section 5.2 of commit `7ec1253`, and those of Sections 5.3 and 5.5 of commit `c5eb6c5`. The semi-synthetic coverage study has its own registration, calibration and result files in `obd_coverage/` and is rerun with `python -m obd_coverage.run`; its results are the output of commit `df8019a`, stored in commit `125c5c2`. This report was built from the result files at commit `9b4c33b`. The first run downloads the dataset archive and the prior file and verifies their SHA-256 checksums; every result file names the commit that produced it. The confirmatory run took 13 minutes. Every number in this report is generated from those result files by a script kept with its source.

Table 12: *Post hoc.* Impressions, complete impressions, the share of complete impressions that show the same list as the one before, and the variance ratio of Section 5.1 for blocks of 1, 10 and 60 minutes. No ratio carries an interval; the 60-minute ratios rest on 84 pairs of blocks.

	Log	Impressions	Complete	Same list	1 min	10 min	60 min
Men's	random	147,487	142,227	37%	1.18	2.45	2.07
	BTS	1,268,311	1,121,295	31%	1.20	2.08	2.02
All	random	423,896	387,225	35%	1.29	1.89	2.10
	BTS	3,731,150	3,040,555	30%	1.14	1.85	2.61
Women's	random	280,211	268,167	36%	1.13	1.92	2.07
	BTS	2,482,614	2,177,760	30%	1.16	2.23	3.33

Table 13: *Post hoc.* Both ratios of H3 by block length: interval, and below it the standard deviation of the replicates with, in brackets, its ratio to the one at ten minutes.

		5 min	10 min	20 min	30 min	60 min	180 min
Men's	IPS	0.790 to 1.001	0.790 to 1.005	0.790 to 1.001	0.789 to 1.000	0.788 to 1.004	0.781 to 1.004
		0.0540 (0.98)	0.0550 (1.00)	0.0538 (0.98)	0.0538 (0.98)	0.0554 (1.01)	0.0567 (1.03)
	SNIPS	0.806 to 1.019	0.805 to 1.022	0.807 to 1.019	0.805 to 1.018	0.804 to 1.022	0.799 to 1.020
		0.0546 (0.98)	0.0556 (1.00)	0.0544 (0.98)	0.0548 (0.98)	0.0560 (1.01)	0.0563 (1.01)
All	IPS	0.870 to 1.028	0.867 to 1.033	0.866 to 1.033	0.863 to 1.037	0.863 to 1.040	0.856 to 1.044
		0.0405 (0.95)	0.0424 (1.00)	0.0433 (1.02)	0.0447 (1.05)	0.0454 (1.07)	0.0481 (1.13)
	SNIPS	0.871 to 1.028	0.867 to 1.032	0.866 to 1.034	0.864 to 1.035	0.864 to 1.039	0.856 to 1.045
		0.0401 (0.95)	0.0421 (1.00)	0.0432 (1.03)	0.0441 (1.05)	0.0449 (1.07)	0.0479 (1.14)
Women's	IPS	0.919 to 1.095	0.917 to 1.099	0.910 to 1.104	0.908 to 1.108	0.906 to 1.112	0.890 to 1.144
		0.0447 (0.96)	0.0467 (1.00)	0.0491 (1.05)	0.0511 (1.09)	0.0524 (1.12)	0.0637 (1.36)
	SNIPS	0.920 to 1.094	0.917 to 1.099	0.910 to 1.102	0.908 to 1.106	0.906 to 1.111	0.890 to 1.140
		0.0445 (0.95)	0.0466 (1.00)	0.0489 (1.05)	0.0510 (1.09)	0.0520 (1.12)	0.0630 (1.35)
Pooled	IPS	0.916 to 1.035	0.915 to 1.036	0.911 to 1.039	0.907 to 1.042	0.909 to 1.042	0.897 to 1.056
		0.0300 (0.97)	0.0310 (1.00)	0.0323 (1.04)	0.0338 (1.09)	0.0343 (1.10)	0.0406 (1.31)
	SNIPS	0.916 to 1.035	0.914 to 1.036	0.912 to 1.039	0.908 to 1.042	0.909 to 1.041	0.896 to 1.057
		0.0298 (0.96)	0.0310 (1.00)	0.0324 (1.05)	0.0335 (1.08)	0.0342 (1.11)	0.0408 (1.32)

Table 14: *Post hoc.* ρ and d of H4 at three hours, the H5 difference and the two ratios of H6 by block length: interval, and below it the standard deviation of the replicates with, in brackets, its ratio to the one at ten minutes. The ten-minute intervals are the registered ones.

		5 min	10 min	20 min	30 min	60 min	180 min
Men's	d	−0.002 to 0.206	−0.010 to 0.190	−0.027 to 0.181	−0.020 to 0.207	−0.018 to 0.192	−0.019 to 0.260
		0.0535 (1.04)	0.0513 (1.00)	0.0528 (1.03)	0.0584 (1.14)	0.0538 (1.05)	0.0713 (1.39)
	ρ	0.862 to 1.122	0.853 to 1.114	0.842 to 1.099	0.848 to 1.120	0.840 to 1.117	0.833 to 1.177
		0.0661 (0.99)	0.0668 (1.00)	0.0663 (0.99)	0.0701 (1.05)	0.0714 (1.07)	0.0881 (1.32)
	H5 difference	0.075 to 0.210	0.072 to 0.218	0.066 to 0.224	0.054 to 0.245	0.010 to 0.304	0.039 to 0.221
		0.0345 (0.93)	0.0371 (1.00)	0.0404 (1.09)	0.0487 (1.31)	0.0750 (2.02)	0.0469 (1.26)
	H6 IPS	0.946 to 1.111	0.942 to 1.114	0.940 to 1.115	0.938 to 1.123	0.940 to 1.121	0.938 to 1.126
		0.0421 (0.96)	0.0439 (1.00)	0.0445 (1.01)	0.0469 (1.07)	0.0455 (1.04)	0.0481 (1.10)
	H6 SNIPS	0.745 to 1.101	0.743 to 1.100	0.742 to 1.101	0.740 to 1.111	0.734 to 1.111	0.741 to 1.117
		0.0951 (1.00)	0.0949 (1.00)	0.0951 (1.00)	0.0979 (1.03)	0.0992 (1.05)	0.1002 (1.06)
All	d	−0.016 to 0.184	−0.019 to 0.182	−0.025 to 0.192	−0.023 to 0.199	−0.025 to 0.186	−0.005 to 0.165
		0.0508 (0.98)	0.0517 (1.00)	0.0558 (1.08)	0.0569 (1.10)	0.0532 (1.03)	0.0436 (0.84)
	ρ	0.932 to 1.134	0.932 to 1.128	0.927 to 1.134	0.925 to 1.145	0.923 to 1.133	0.934 to 1.126
		0.0511 (1.02)	0.0503 (1.00)	0.0525 (1.04)	0.0558 (1.11)	0.0539 (1.07)	0.0486 (0.97)
	H5 difference	0.035 to 0.165	0.033 to 0.166	0.025 to 0.171	0.012 to 0.184	−0.022 to 0.220	0.023 to 0.169
		0.0333 (0.98)	0.0341 (1.00)	0.0367 (1.08)	0.0432 (1.27)	0.0611 (1.79)	0.0368 (1.08)
	H6 IPS	0.920 to 1.068	0.918 to 1.070	0.917 to 1.072	0.915 to 1.077	0.916 to 1.074	0.919 to 1.066
		0.0377 (0.96)	0.0392 (1.00)	0.0393 (1.00)	0.0412 (1.05)	0.0401 (1.02)	0.0377 (0.96)
	H6 SNIPS	0.858 to 1.043	0.855 to 1.050	0.856 to 1.051	0.854 to 1.053	0.855 to 1.052	0.858 to 1.045
		0.0470 (0.95)	0.0494 (1.00)	0.0497 (1.00)	0.0509 (1.03)	0.0502 (1.02)	0.0480 (0.97)
Women's	d	−0.031 to 0.144	−0.031 to 0.147	−0.032 to 0.155	−0.029 to 0.156	−0.033 to 0.168	−0.035 to 0.144
		0.0447 (0.98)	0.0455 (1.00)	0.0479 (1.05)	0.0473 (1.04)	0.0519 (1.14)	0.0456 (1.00)
	ρ	0.975 to 1.151	0.974 to 1.158	0.969 to 1.168	0.967 to 1.175	0.960 to 1.190	0.959 to 1.180
		0.0453 (0.96)	0.0471 (1.00)	0.0503 (1.07)	0.0530 (1.13)	0.0584 (1.24)	0.0559 (1.19)
	H5 difference	0.025 to 0.129	0.022 to 0.130	0.015 to 0.138	−0.001 to 0.153	−0.039 to 0.196	−0.005 to 0.140
		0.0262 (0.94)	0.0278 (1.00)	0.0319 (1.15)	0.0391 (1.41)	0.0594 (2.14)	0.0369 (1.33)
	H6 IPS	0.899 to 1.017	0.897 to 1.019	0.893 to 1.023	0.896 to 1.021	0.890 to 1.028	0.885 to 1.027
		0.0303 (0.97)	0.0312 (1.00)	0.0335 (1.07)	0.0319 (1.02)	0.0351 (1.12)	0.0366 (1.17)
	H6 SNIPS	0.821 to 1.013	0.819 to 1.014	0.818 to 1.021	0.817 to 1.018	0.818 to 1.017	0.825 to 1.016
		0.0490 (0.97)	0.0504 (1.00)	0.0515 (1.02)	0.0511 (1.01)	0.0508 (1.01)	0.0493 (0.98)
Pooled	d	0.003 to 0.137	0.002 to 0.139	0.001 to 0.144	0.001 to 0.150	0.001 to 0.147	0.004 to 0.132
		0.0343 (0.98)	0.0349 (1.00)	0.0367 (1.05)	0.0380 (1.09)	0.0370 (1.06)	0.0330 (0.95)
	ρ	0.976 to 1.113	0.978 to 1.114	0.976 to 1.116	0.973 to 1.127	0.970 to 1.124	0.972 to 1.116
		0.0348 (1.00)	0.0348 (1.00)	0.0359 (1.03)	0.0387 (1.11)	0.0394 (1.13)	0.0365 (1.05)

Table 15: *Post hoc.* The first hour of the week by position: rows, the registered distance between the hour's frequencies and the target, the floor from the alternate ten-minute blocks of that hour, and the distance over the floor. The target is the prior for the BTS log and the uniform distribution for the random log.

	Log	Position	Rows	Distance	Floor	Distance / floor
Men's	BTS	1	8,473	0.149	0.123	1.21
		2	8,480	0.096	0.107	0.89
		3	8,480	0.067	0.102	0.66
	random	1	961	0.174	0.313	0.56
		2	963	0.214	0.356	0.60
		3	964	0.195	0.350	0.56
All	BTS	1	25,432	0.180	0.138	1.30
		2	25,438	0.126	0.153	0.83
		3	25,445	0.088	0.150	0.59
	random	1	2,822	0.166	0.345	0.48
		2	2,828	0.165	0.434	0.38
		3	2,822	0.191	0.373	0.51
Women's	BTS	1	16,059	0.164	0.116	1.42
		2	16,065	0.098	0.118	0.83
		3	16,060	0.105	0.099	1.06
	random	1	1,778	0.165	0.275	0.60
		2	1,774	0.194	0.262	0.74
		3	1,774	0.158	0.331	0.48

Table 16: H7 as registered (first column) and, post hoc (second column), with the two values of the unmatched feature taken as one. An estimate of a distance can lie below its own interval, because a distance between two resampled logs is biased upward.

	Registered	Post hoc
All		
Distance, the unmatched feature	0.355 (0.352 to 0.358)	0.002 (0.001 to 0.004)
Joint distance	0.360 (0.359 to 0.365)	0.007 (0.008 to 0.011)
BTS rows with a combination absent from the random log	0.353 (0.350 to 0.356)	0.0001 (0.0001 to 0.0002)
Change in ρ under user weights	−0.058 (−0.136 to 0.013)	0.001 (−0.003 to 0.005)
Women's		
Distance, the unmatched feature	0.228 (0.227 to 0.231)	0.002 (0.001 to 0.004)
Joint distance	0.236 (0.236 to 0.241)	0.009 (0.010 to 0.014)
BTS rows with a combination absent from the random log	0.228 (0.226 to 0.230)	0.0001 (0.0002 to 0.0003)
Change in ρ under user weights	−0.116 (−0.178 to −0.063)	−0.001 (−0.014 to 0.005)

Table 17: *Post hoc.* M of H2 at the registered eligibility bound and two lower ones, the first-order grid bias, and the part of the distance of S from 1 that falls in hours with an hourly S under 0.1 and under 0.5, averaged over the eligible cells. Below the registered bound the last two columns stop measuring a bias: at a propensity of 0.0001 a cell is expected to have hours without a row.

	Bound	Cells	M	Grid bias	Under 0.1	Under 0.5
Men's	0.001	12	1.005 (0.997 to 1.013)	+0.0002	0.002	0.007
	0.0005	25	0.996 (0.986 to 1.007)	+0.0005	0.018	0.041
	0.0001	90	0.973 (0.961 to 0.986)	+0.0028	0.134	0.179
All	0.001	3	0.973 (0.957 to 0.990)	+0.0002	0.033	0.039
	0.0005	16	0.984 (0.973 to 0.996)	+0.0006	0.026	0.048
	0.0001	135	0.950 (0.938 to 0.962)	+0.0040	0.161	0.207
Women's	0.001	7	0.967 (0.955 to 0.980)	+0.0002	0.022	0.034
	0.0005	22	0.978 (0.968 to 0.989)	+0.0005	0.030	0.049
	0.0001	99	0.956 (0.944 to 0.968)	+0.0030	0.134	0.176

Table 18: *Post hoc.* M of H2 at the registered bound by UTC day, from the blocks of that day.

	Men's	All	Women's
24 November	1.010 (0.990 to 1.030)	1.022 (0.988 to 1.058)	0.974 (0.945 to 1.003)
25 November	1.004 (0.968 to 1.042)	1.015 (0.977 to 1.055)	0.993 (0.956 to 1.030)
26 November	0.994 (0.977 to 1.010)	1.023 (0.987 to 1.060)	0.995 (0.973 to 1.017)
27 November	1.004 (0.987 to 1.021)	0.879 (0.832 to 0.931)	0.946 (0.922 to 0.973)
28 November	1.006 (0.984 to 1.029)	0.981 (0.946 to 1.016)	0.999 (0.971 to 1.028)
29 November	1.006 (0.984 to 1.029)	0.924 (0.881 to 0.968)	0.863 (0.816 to 0.908)
30 November	1.007 (0.989 to 1.026)	0.982 (0.932 to 1.034)	1.003 (0.976 to 1.031)